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### METHOD FOR DETERMINING POINT IN TIME FOR APPLYING CHANGED MINIMAL SCHEDULING OFFSET IN WIRELESS COMMUNICATION SYSTEM, AND DEVICE FOR APPLYING METHOD

#### Abstract

Provided are a method for determining a point in time for applying a changed minimal scheduling offset in a wireless communication system, and a device for applying the method. The method comprises the steps of: receiving, in slot  $n$  of a scheduling cell, first DCI comprising information for informing a change in a  $K \cdot 2 \text{ min}$  value which is a minimal scheduling offset; receiving second DCI based on the changed  $K \cdot 2 \text{ min}$  value after slot  $n+X$  of the scheduling cell; and transmitting a PUSCH scheduled by the second DCI. The value of  $X$  is determined on the basis of:  $K \cdot 0 \text{ min}$  which is a minimal scheduling offset currently applied to a scheduled cell scheduled by the first DCI; a subcarrier interval setting of the scheduling cell; a subcarrier interval setting of the scheduled cell; a value predetermined dependently on a subcarrier interval of the scheduling cell; etc.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 17/774,403, filed on May 4, 2022, which is a National Stage application under 35 U.S.C. § 371 of International Application No. PCT/KR 2020/015134, filed on Nov. 2, 2020, which claims the benefit of Korean Patent Application No. 10-2019-0142077, filed on Nov. 7, 2019. The disclosures of the prior applications are hereby incorporated by reference herein in their entirety.

### BACKGROUNDS

#### Field of the Description

[0002] The present disclosure relates to a method for determining when to apply a changed minimum scheduling offset in a wireless communication system, and an apparatus using the method.

#### Related Art

[0003] As a growing number of communication devices require higher communication capacity, there is a need for advanced mobile broadband communication as compared to existing radio access technology (RAT). Massive machine-type communication (MTC), which provides a variety of services anytime and anywhere by connecting a plurality of devices and a plurality of objects, is also one major issue to be considered in next-generation communication. In addition, designs for communication systems considering services or a user equipment (UE) sensitive to reliability and latency are under discussion. Introduction of next-generation RAT considering enhanced mobile broadband communication, massive MTC, and ultra-reliable and low-latency communication (URLLC) is under discussion. In this disclosure, for convenience of description, this technology may be referred to as new RAT or new radio (NR). NR is also referred to as a fifth generation (5G) system.

[0004] As the performance and functions of the UE such as display resolution, display size, processor, memory, and application increase of the UE improve, power consumption also increases. Since the power supply of the UE may be limited to the battery, it is important to reduce power consumption. This is the same for a UE operating in NR.

[0005] As one example for reducing power consumption of the UE, there is cross-slot scheduling. A slot in which a physical downlink control channel (PDCCH) is received is the same as a slot in which a physical downlink shared channel (PDSCH) scheduled by the PDCCH is received is called same-slot scheduling. In cross-slot scheduling, the PDCCH and the PDSCH scheduled by the PDCCH may be in different slots. The PDCCH is often received only in some symbols within the slot (e.g., the first 3 symbols of the slot) and decoded. When cross-slot scheduling is applied, the UE can save power by making a radio frequency (RF) unit sleep in symbols (slots) after receiving the PDCCH before receiving the PDSCH.

[0006] The network sets the minimum applicable K 0/K 2 value, and may indicate to the UE the minimum applicable slot offset between the PDCCH (more specifically, downlink control information (DCI)) and the PDSCH/PUSCH (physical uplink shared channel) scheduled by the

DCI. The minimum applicable slot offset means a minimum value of an offset (difference) between a slot in which DCI is received and a PDSCH/PUSCH slot scheduled by the DCI, and may be referred to as a minimum scheduling offset.

[0007] By the way, upon indicating/changing the minimum scheduling offset, when the indication/change is applied may be defined through an 'application delay' value.

[0008] When cross-carrier scheduling, particularly, cross-carrier scheduling is used in carrier aggregation in which the numerology of the scheduling cell and the scheduled cell are different, it is necessary to clearly define how to determine the application delay value. Otherwise, it may be difficult to apply the power saving technique of the UE.

## SUMMARY

[0009] The technical problem to be solved by the present disclosure is to provide a method for determining when to apply a changed minimum scheduling offset in a wireless communication system, and an apparatus using the method.

[0010] In one aspect, provided is a method for determining when to apply a changed minimum scheduling offset in a wireless communication system. The method includes receiving first downlink control information (DCI) including information for a change to a value of  $K \cdot 2 \text{ min}$ , which is a minimum scheduling offset, in a slot  $n$  of a scheduling cell, receiving second DCI based on a changed  $K \cdot 2 \text{ min}$  value from slot  $n+X$  of the scheduling cell, and transmitting a physical uplink shared channel (PUSCH) scheduled by the second DCI. The  $X$  value is a largest value among i) a first value obtained by multiplying currently applied minimum scheduling offset,  $K \cdot 0 \text{ min}$  ( $Y$ ), in a scheduled cell scheduled by the first DCI by  $2 \cdot \text{sup.}\mu\text{scheduling}/2 \cdot \text{sup.}\mu\text{scheduled}$  and then performing ceiling and ii) a second value ( $Z$ ) that are predetermined depending on a subcarrier spacing (SCS) of the scheduling cell, and the  $\mu\text{scheduling}$  is a subcarrier spacing configuration of the scheduling cell and the  $\mu\text{scheduled}$  is a subcarrier spacing configuration of the scheduled cell.

[0011] In another aspect, provided is a user equipment (UE). The UE includes a transceiver for transmitting and receiving a radio signal and a processor operating in connected to the transceiver. The processor is configured to: receive first downlink control information (DCI) including information for a change to a value of  $K \cdot 2 \text{ min}$ , which is a minimum scheduling offset, in a slot  $n$  of a scheduling cell, receive second DCI based on a changed  $K \cdot 2 \text{ min}$  value from slot  $n+X$  of the scheduling cell, and transmit a physical uplink shared channel (PUSCH) scheduled by the second DCI. The  $X$  value is a largest value among i) a first value obtained by multiplying currently applied minimum scheduling offset,  $K \cdot 0 \text{ min}$  ( $Y$ ), in a scheduled cell scheduled by the first DCI by  $2 \cdot \text{sup.}\mu\text{scheduling}/2 \cdot \text{sup.}\mu\text{scheduled}$  and then performing ceiling and ii) a second value ( $Z$ ) that are predetermined depending on a subcarrier spacing (SCS) of the scheduling cell, and the  $\mu\text{scheduling}$  is a subcarrier spacing configuration of the scheduling cell and the  $\mu\text{scheduled}$  is a subcarrier spacing configuration of the scheduled cell.

[0012] In still another aspect, provided is a method of operating a base station in a wireless communication system. The method includes transmitting, to a user equipment (UE), first downlink control information (DCI) including information for a change to a value of  $K \cdot 2 \text{ min}$ , which is a minimum scheduling offset, in a slot  $n$  of a scheduling cell, transmitting, to the UE, second DCI based on a changed  $K \cdot 2 \text{ min}$  value from slot  $n+X$  of the scheduling cell, and receiving, from the UE, a physical uplink shared channel (PUSCH) scheduled by the second DCI. The  $X$  value is a largest value among i) a first value obtained by multiplying currently applied minimum scheduling offset,  $K \cdot 0 \text{ min}$  ( $Y$ ), in a scheduled cell scheduled by the first DCI by  $2 \cdot \text{sup.}\mu\text{scheduling}/2 \cdot \text{sup.}\mu\text{scheduled}$  and then performing ceiling and ii) a second value ( $Z$ ) that are predetermined depending on a subcarrier spacing (SCS) of the scheduling cell, and the  $\mu\text{scheduling}$  is a subcarrier spacing configuration of the scheduling cell and the  $\mu\text{scheduled}$  is a subcarrier spacing configuration of the scheduled cell.

[0013] In still another aspect, provided is a base station. The base station includes a transceiver for transmitting and receiving a radio signal and a processor operating in connected to the transceiver.

The processor is configured to: transmit, to a user equipment (UE), first downlink control information (DCI) including information for a change to a value of  $K \cdot 2 \text{ min}$ , which is a minimum scheduling offset, in a slot  $n$  of a scheduling cell, transmit, to the UE, second DCI based on a changed  $K \cdot 2 \text{ min}$  value from slot  $n+X$  of the scheduling cell, and receive, from the UE, a physical uplink shared channel (PUSCH) scheduled by the second DCI. The  $X$  value is a largest value among i) a first value obtained by multiplying currently applied minimum scheduling offset,  $K \cdot 0 \text{ min}$  ( $Y$ ), in a scheduled cell scheduled by the first DCI by  $2 \cdot \mu_{\text{scheduling}} / 2 \cdot \mu_{\text{scheduled}}$  and then performing ceiling and ii) a second value ( $Z$ ) that are predetermined depending on a subcarrier spacing (SCS) of the scheduling cell, and the  $\mu_{\text{scheduling}}$  is a subcarrier spacing configuration of the scheduling cell and the  $\mu_{\text{scheduled}}$  is a subcarrier spacing configuration of the scheduled cell.

[0014] In still another aspect, provided is at least one computer-readable medium (CRM) comprising an instruction based on being executed by at least one processor. The CRM receives first downlink control information (DCI) including information for a change to a value of  $K \cdot 2 \text{ min}$ , which is a minimum scheduling offset, in a slot  $n$  of a scheduling cell, receives second DCI based on a changed  $K \cdot 2 \text{ min}$  value from slot  $n+X$  of the scheduling cell, and transmits a physical uplink shared channel (PUSCH) scheduled by the second DCI. The  $X$  value is a largest value among i) a first value obtained by multiplying currently applied minimum scheduling offset,  $K \cdot 0 \text{ min}$  ( $Y$ ), in a scheduled cell scheduled by the first DCI by  $2 \cdot \mu_{\text{scheduling}} / 2 \cdot \mu_{\text{scheduled}}$  and then performing ceiling and ii) a second value ( $Z$ ) that are predetermined depending on a subcarrier spacing (SCS) of the scheduling cell, and the  $\mu_{\text{scheduling}}$  is a subcarrier spacing configuration of the scheduling cell and the  $\mu_{\text{scheduled}}$  is a subcarrier spacing configuration of the scheduled cell.

[0015] In still another aspect, provided is an apparatus operated in a wireless communication system. The apparatus includes a processor and a memory to be operatively connected to the processor. The processor is configured to: receive first downlink control information (DCI) including information for a change to a value of  $K \cdot 2 \text{ min}$ , which is a minimum scheduling offset, in a slot  $n$  of a scheduling cell, receive second DCI based on a changed  $K \cdot 2 \text{ min}$  value from slot  $n+X$  of the scheduling cell, and transmit a physical uplink shared channel (PUSCH) scheduled by the second DCI. The  $X$  value is a largest value among i) a first value obtained by multiplying currently applied minimum scheduling offset,  $K \cdot 0 \text{ min}$  ( $Y$ ), in a scheduled cell scheduled by the first DCI by  $2 \cdot \mu_{\text{scheduling}} / 2 \cdot \mu_{\text{scheduled}}$  and then performing ceiling and ii) a second value ( $Z$ ) that are predetermined depending on a subcarrier spacing (SCS) of the scheduling cell, and the  $\mu_{\text{scheduling}}$  is a subcarrier spacing configuration of the scheduling cell and the  $\mu_{\text{scheduled}}$  is a subcarrier spacing configuration of the scheduled cell.

[0016] When a change in the minimum scheduling offset is indicated in carrier aggregation using different numerologies (e.g., different subcarrier spacing) in the scheduling cell and the scheduled cell, misunderstandings do not occur between the network and the UE by clarifying the application delay value indicating the time of application of the change. As a result, the UE can accurately know the slot/symbol in which the UE can perform a power saving operation (e.g., a sleep operation), thereby increasing communication efficiency. In addition, it prevents an impossible or difficult UE operation from occurring by determining the application delay value in consideration of the position in the slot of the DCI indicating the change.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0017] FIG. 1 shows a wireless communication system to which the present disclosure may be applied.

[0018] FIG. 2 is a diagram showing a wireless protocol architecture for a user plane.

[0019] FIG. 3 is a diagram showing a wireless protocol architecture for a control plane.

[0020] FIG. 4 shows another example of a wireless communication system to which the present disclosure may be applied.

[0021] FIG. 5 illustrates a functional division between an NG-RAN and a 5GC.

[0022] FIG. 6 illustrates an example of a frame structure that may be applied in NR.

[0023] FIG. 7 illustrates a slot structure of the NR frame.

[0024] FIG. 8 illustrates a CORESET.

[0025] FIG. 9 illustrates a structure of self-contained slot.

[0026] FIG. 10 shows an example of a downlink transmission/reception operation.

[0027] FIG. 11 shows an example of an uplink transmission/reception operation.

[0028] FIG. 12 shows an example of an uplink grant.

[0029] FIG. 13 shows an example of a grant-free initial transmission.

[0030] FIG. 14 is an example of applying an application delay.

[0031] FIG. 15 illustrates a location of a CORESET for PDCCH monitoring.

[0032] FIG. 16 illustrates a method for determining an application delay value according to option 3.

[0033] FIG. 17 is an example of applying the method of FIG. 16.

[0034] FIG. 18 is another example of applying the method of FIG. 16.

[0035] FIG. 19 illustrates a signaling method between a network (base station) and a UE.

[0036] FIG. 20 illustrates a signaling method between a network (base station) and a UE.

[0037] FIG. 21 illustrates a wireless device that is applicable to the disclosure.

[0038] FIG. 22 illustrates a signal processing circuit for a transmission signal.

[0039] FIG. 23 shows another example of the structure of a signal processing module in a transmission device.

[0040] FIG. 24 illustrates an example of a wireless communication device according to an implementation of the present disclosure.

[0041] FIG. 25 shows an example of a processor **2000**.

[0042] FIG. 26 shows an example of a processor **3000**.

[0043] FIG. 27 shows another example of a wireless device.

[0044] FIG. 28 shows another example of a wireless device applied to the present specification.

[0045] FIG. 29 illustrates a portable device applied to the present specification.

[0046] FIG. 30 illustrates the communication system **1** applied to this specification.

[0047] FIG. 31 illustrates a vehicle or autonomous driving vehicle that may be applied herein.

#### DESCRIPTION OF EXEMPLARY EMBODIMENTS

[0048] FIG. 1 shows a wireless communication system to which the present disclosure may be applied. The wireless communication system may be referred to as an Evolved-UMTS Terrestrial Radio Access Network (E-UTRAN) or a Long Term Evolution (LTE)/LTE-A system.

[0049] The E-UTRAN includes at least one base station (BS) **20** which provides a control plane and a user plane to a user equipment (UE) **10**. The UE **10** may be fixed or mobile, and may be referred to as another terminology, such as a mobile station (MS), a user terminal (UT), a subscriber station (SS), a mobile terminal (MT), a wireless device, terminal, etc. The BS **20** is generally a fixed station that communicates with the UE **10** and may be referred to as another terminology, such as an evolved node-B (eNB), a base transceiver system (BTS), an access point, gN B, etc.

[0050] The BSs **20** are interconnected by means of an X2 interface. The BSs **20** are also connected by means of an S1 interface to an evolved packet core (EPC) **30**, more specifically, to a mobility management entity (MME) through S1-MME and to a serving gateway (S-GW) through S1-U.

[0051] The EPC **30** includes an MME, an S-GW, and a packet data network-gateway (P-GW). The MME has access information of the UE or capability information of the UE, and such information is generally used for mobility management of the UE. The S-GW is a gateway having an E-UTRAN as an end point. The P-GW is a gateway having a PDN as an end point.

[0052] Layers of a radio interface protocol between the UE and the network can be classified into a first layer (L1), a second layer (L2), and a third layer (L3) based on the lower three layers of the open system interconnection (OSI) model that is well-known in the communication system. Among them, a physical (PHY) layer belonging to the first layer provides an information transfer service by using a physical channel, and a radio resource control (RRC) layer belonging to the third layer serves to control a radio resource between the UE and the network. For this, the RRC layer exchanges an RRC message between the UE and the BS.

[0053] FIG. 2 is a diagram showing a wireless protocol architecture for a user plane. FIG. 3 is a diagram showing a wireless protocol architecture for a control plane. The user plane is a protocol stack for user data transmission. The control plane is a protocol stack for control signal transmission.

[0054] Referring to FIGS. 2 and 3, a PHY layer provides a higher layer with an information transfer service through a physical channel. The PHY layer is connected to a medium access control (MAC) layer which is a higher layer of the PHY layer through a transport channel. Data is transferred between the MAC layer and the PHY layer through the transport channel. The transport channel is classified according to how and with what characteristics data is transferred through a radio interface.

[0055] Data is moved between different PHY layers, that is, the PHY layers of a transmitter and a receiver, through a physical channel. The physical channel may be modulated according to an Orthogonal Frequency Division Multiplexing (OFDM) scheme, and use the time and frequency as radio resources.

[0056] The functions of the MAC layer include mapping between a logical channel and a transport channel and multiplexing and demultiplexing to a transport block that is provided through a physical channel on the transport channel of a MAC Service Data Unit (SDU) that belongs to a logical channel. The MAC layer provides service to a Radio Link Control (RLC) layer through the logical channel.

[0057] The functions of the RLC layer include the concatenation, segmentation, and reassembly of an RLC SDU. In order to guarantee various types of Quality of Service (QoS) required by a Radio Bearer (RB), the RLC layer provides three types of operation mode: Transparent Mode (TM), Unacknowledged Mode (UM), and Acknowledged Mode (AM). AM RLC provides error correction through an Automatic Repeat Request (ARQ).

[0058] The RRC layer is defined only on the control plane. The RRC layer is related to the configuration, reconfiguration, and release of radio bearers, and is responsible for control of logical channels, transport channels, and PHY channels. An RB means a logical route that is provided by the first layer (PHY layer) and the second layers (MAC layer, the RLC layer, and the PDCP layer) in order to transfer data between UE and a network.

[0059] The function of a Packet Data Convergence Protocol (PDCP) layer on the user plane includes the transfer of user data and header compression and ciphering. The function of the PDCP layer on the user plane further includes the transfer and encryption/integrity protection of control plane data.

[0060] What an RB is configured means a process of defining the characteristics of a wireless protocol layer and channels in order to provide specific service and configuring each detailed parameter and operating method. An RB can be divided into two types of a Signaling RB (SRB) and a Data RB (DRB). The SRB is used as a passage through which an RRC message is transmitted on the control plane, and the DRB is used as a passage through which user data is transmitted on the user plane.

[0061] If RRC connection is established between the RRC layer of UE and the RRC layer of an E-UTRAN, the UE is in the RRC connected state. If not, the UE is in the RRC idle state.

[0062] A downlink transport channel through which data is transmitted from a network to UE includes a broadcast channel (BCH) through which system information is transmitted and a

downlink shared channel (SCH) through which user traffic or control messages are transmitted. Traffic or a control message for downlink multicast or broadcast service may be transmitted through the downlink SCH, or may be transmitted through an additional downlink multicast channel (MCH). Meanwhile, an uplink transport channel through which data is transmitted from UE to a network includes a random access channel (RACH) through which an initial control message is transmitted and an uplink shared channel (SCH) through which user traffic or control messages are transmitted.

[0063] Logical channels that are placed over the transport channel and that are mapped to the transport channel include a broadcast control channel (BCCH), a paging control channel (PCCH), a common control channel (CCCH), a multicast control channel (M CCH), and a multicast traffic channel (MTCH).

[0064] The physical channel includes several OFDM symbols in the time domain and several subcarriers in the frequency domain. One subframe includes a plurality of OFDM symbols in the time domain. An RB is a resources allocation unit, and includes a plurality of OFDM symbols and a plurality of subcarriers. Furthermore, each subframe may use specific subcarriers of specific OFDM symbols (e.g., the first OFDM symbol) of the corresponding subframe for a physical downlink control channel (PDCCH), that is, an L1/L2 control channel. A Transmission Time Interval (TTI) is a unit time for transmission.

[0065] Hereinafter, a new radio access technology (new RAT, NR) will be described.

[0066] As more and more communication devices require more communication capacity, there is a need for improved mobile broadband communication over existing radio access technology. A Iso, massive machine type communications (MTC), which provides various services by connecting many devices and objects, is one of the major issues to be considered in the next generation communication. In addition, communication system design considering reliability/latency sensitive service/UE is being discussed. The introduction of next generation radio access technology considering enhanced mobile broadband communication (eMBB), massive MTC (mMTC), ultrareliable and low latency communication (URLLC) is discussed. This new technology may be called new radio access technology (new RAT or NR) in the present disclosure for convenience.

[0067] FIG. 4 illustrates a system structure of a next generation radio access network (NG-RAN) to which NR is applied.

[0068] Referring to FIG. 4, the NG-RAN may include a gNB 21 and/or an ng-eNB 22 that provides user plane and control plane protocol termination to a UE. The gNB 21 and the ng-eNB 22 are connected by an Xn interface. The gNB 21 and the ng-eNB 22 are connected to a 5G core network (5GC) via an NG interface. More specifically, the gNB and the eNB are connected to an access and mobility management function (AMF) 31 via an NG-C interface and connected to a user plane function (UPF) 31 via an NG-U interface.

[0069] FIG. 5 illustrates a functional division between an NG-RAN and a 5GC.

[0070] The gNB may provide functions such as an inter-cell radio resource management (Inter Cell RRM), radio bearer management (RB control), connection mobility control, radio admission control, measurement configuration & provision, dynamic resource allocation, and the like. The AMF may provide functions such as NAS security, idle state mobility handling, and so on. The UPF may provide functions such as mobility anchoring, PDU processing, and the like. The SM F may provide functions such as UE IP address assignment, PDU session control, and so on.

[0071] FIG. 6 illustrates an example of a frame structure that may be applied in NR.

[0072] Referring to FIG. 6, a radio frame (which may be called as a frame hereinafter) may be used for uplink and downlink transmission in NR. A frame has a length of 10 ms and may be defined as two 5 ms half-frames (Half-Frame, HF). A half-frame may be defined as five 1 ms subframes (Subframe, SF). A subframe may be divided into one or more slots, and the number of slots in a subframe depends on subcarrier spacing (SCS). Each slot includes 12 or 14 OFDM (A) symbols according to a cyclic prefix (CP). When a normal CP is used, each slot includes 14 symbols. When

an extended CP is used, each slot includes 12 symbols. Here, the symbol may include an OFDM symbol (or a CP-OFDM symbol) and an SC-FDMA symbol (or a DFT-s-OFDM symbol).

[0073] The following table 1 illustrates a subcarrier spacing configuration  $\mu$ .

[Table 1]

TABLE-US-00001 TABLE 1  $\mu$   $\Delta f = 2^{\mu} \cdot 15$  [kHz] Cyclic prefix 0 15 Normal 1 30 Normal 2 60 Normal Extended 3 120 Normal 4 240 Normal

[0074] The following table 2 illustrates the number of slots in a frame ( $N_{\text{slot}}^{\text{frame}, \mu}$ ), the number of slots in a subframe ( $N_{\text{slot}}^{\text{subframe}, \mu}$ ), the number of symbols in a slot ( $N_{\text{slot}}^{\text{sub.symb}, \mu}$ ), and the like, according to subcarrier spacing configurations  $\mu$ .

[Table 2]

TABLE-US-00002 TABLE 2  $\mu$   $N_{\text{slot}}^{\text{sub.symb}, \mu}$   $N_{\text{slot}}^{\text{sub.slot}, \mu}$   $N_{\text{slot}}^{\text{sub.frame}, \mu}$   $N_{\text{slot}}^{\text{sub.subframe}, \mu}$  0 14 10 1 1 14 20 2 2 14 40 4 3 14 80 8 4 14 160 16

[0075] In FIG. 6,  $\mu=0, 1, 2$ , and 3 are exemplified.

[0076] Table 2-1 below exemplifies that the number of symbols per slot, the number of slots per frame, and the number of slots per subframe vary according to SCS ( $\mu=2, 60$  kHz) when the extended CP is used.

[Table 2-1]

TABLE-US-00003 TABLE 2-1  $\mu$   $N_{\text{slot}}^{\text{sub.symb}, \mu}$   $N_{\text{slot}}^{\text{sub.slot}, \mu}$   $N_{\text{slot}}^{\text{sub.frame}, \mu}$   $N_{\text{slot}}^{\text{sub.subframe}, \mu}$  2 12 40 4

[0077] In an NR system, OFDM (A) numerologies (e.g., SCS, CP length, and so on) may be differently configured between a plurality of cells integrated to one UE. Accordingly, an (absolute time) duration of a time resource (e.g., SF, slot or TTI) (for convenience, collectively referred to as a time unit (TU)) configured of the same number of symbols may be differently configured between the integrated cells.

[0078] FIG. 7 illustrates a slot structure of a NR frame.

[0079] A slot may comprise a plurality of symbols in a time domain. For example, in case of a normal CP, one slot may include 7 symbols. However, in case of an extended CP, one slot may include 6 symbols. The carrier may include a plurality of subcarriers in a frequency domain. A resource block (RB) may be defined as a plurality of consecutive subcarriers (e.g., 12) in the frequency domain. A bandwidth part (BWP) may be defined as a plurality of consecutive (P)RBs in the frequency domain, and may correspond to one numerology (e.g., SCS, CP length, etc.). A carrier may include a maximum of N (e.g., 5) BWPs. Data communication is performed through the activated BWP, and only one BWP can be activated for one UE. Each element in the resource grid is referred to as a resource element (RE), and one complex symbol may be mapped thereto.

[0080] A physical downlink control channel (PDCCH) may include one or more control channel elements (CCEs) as illustrated in the following table 3.

[Table 3]

TABLE-US-00004 TABLE 3 Aggregation level Number of CCEs 1 1 2 2 4 4 8 8 16 16

[0081] That is, the PDCCH may be transmitted through a resource including 1, 2, 4, 8, or 16 CCEs. Here, the CCE includes six resource element groups (REGs), and one REG includes one resource block in a frequency domain and one orthogonal frequency division multiplexing (OFDM) symbol in a time domain.

[0082] Monitoring means decoding each PDCCH candidate according to a downlink control information (DCI) format. The UE monitors a set of PDCCH candidates in one or more CORESETs (described below) on the activated DL BWP of each activated serving cell for which PDCCH monitoring is configured according to a corresponding search space set.

[0083] A new unit called a control resource set (CORESET) may be introduced in the NR. The UE may receive a PDCCH in the CORESET.

[0084] FIG. 8 illustrates a CORESET.

[0085] Referring to FIG. 8, the CORESET includes  $N_{\text{CORESET}}^{\text{sub.RB}}$  resource blocks in

the frequency domain, and  $N_{\text{sup.CORESET.sub.symb}} \in \{1, 2, 3\}$  number of symbols in the time domain. FIG. 8 shows an example in which the CORESET is composed of two symbols in the time domain.  $N_{\text{sup.CORESET.sub.RB}}$  and  $N_{\text{sup.CORESET.sub.symb}}$  may be provided by a base station via higher layer signaling. A plurality of CCEs (or REGs) may be included in the CORESET. One CCE may be composed of a plurality of resource element groups (REGs), and one REG may include one OFDM symbol in the time domain and 12 resource elements in the frequency domain.

[0086] The UE may attempt to detect a PDCCH in units of 1, 2, 4, 8, or 16 CCEs in the CORESET. One or a plurality of CCEs in which PDCCH detection may be attempted may be referred to as PDCCH candidates.

[0087] A plurality of CORESETs may be configured for the UE.

[0088] A control region in the conventional wireless communication system (e.g., LTE/LTE-A) is configured over the entire system band used by a base station (BS). All the UEs, excluding some (e.g., eMTC/NB-IoT UE) supporting only a narrow band, must be able to receive wireless signals of the entire system band of the BS in order to properly receive/decode control information transmitted by the BS.

[0089] On the other hand, in NR, CORESET described above was introduced. CORESET is radio resources for control information to be received by the UE and may use only a portion, rather than the entirety of the system bandwidth in the frequency domain. In addition, in the time domain, only some of the symbols in the slot may be used. The BS may allocate the CORESET to each UE and may transmit control information through the allocated CORESET. For example, a first CORESET may be allocated to UE 1, a second CORESET may be allocated to UE 2, and a third CORESET may be allocated to UE 3. In the NR, the UE may receive control information from the BS, without necessarily receiving the entire system band.

[0090] The CORESET may include a UE-specific CORESET for transmitting UE-specific control information and a common CORESET for transmitting control information common to all UEs.

[0091] Meanwhile, NR may require high reliability according to applications. In such a situation, a target block error rate (BLER) for downlink control information (DCI) transmitted through a downlink control channel (e.g., physical downlink control channel (PDCCH)) may remarkably decrease compared to those of conventional technologies. As an example of a method for satisfying requirement that requires high reliability, content included in DCI can be reduced and/or the amount of resources used for DCI transmission can be increased. Here, resources can include at least one of resources in the time domain, resources in the frequency domain, resources in the code domain and resources in the spatial domain.

[0092] In NR, the following technologies/features can be applied.

<Self-Contained Subframe Structure>

[0093] In NR, a structure in which a control channel and a data channel are time-division-multiplexed (TDM) within one TTI can be considered as a frame structure in order to minimize latency. This structure is characterized in that DL transmission and UL transmission are sequentially performed within one subframe (or slot, hereinafter the same) and thus DL data can be transmitted and UL ACK/NACK can be received within the subframe. Consequently, a time required from occurrence of a data transmission error to data retransmission is reduced, thereby minimizing latency in final data transmission.

[0094] In this data and control TDM ed subframe structure, a time gap for a base station and a UE to switch from a transmission mode to a reception mode or from the reception mode to the transmission mode may be required. To this end, some OFDM symbols at a time when DL switches to UL may be set to a guard period (GP) in the self-contained subframe structure.

[0095] FIG. 9 illustrates a structure of self-contained slot.

[0096] In NR system, one slot includes all of a DL control channel, DL or UL data channel, UL control channel, and so on. For example, the first N symbols in a slot may be used for transmitting

a DL control channel (in what follows, DL control region), and the last M symbols in the slot may be used for transmitting an UL control channel (in what follows, UL control region). N and M are each an integer of 0 or larger. A resource region located between the DL and UL control regions (in what follows, a data region) may be used for transmission of DL data or UL data. As one example, one slot may correspond to one of the following configurations. Each period is listed in the time order. [0097] 1. DL only configuration [0098] 2. UL only configuration [0099] 3. Mixed UL-DL configuration [0100] DL region+GP (Guard Period)+UL control region [0101] DL control region+GP+UL region [0102] a DL region: (i) a DL data region, (ii) DL control region plus DL data region [0103] a UL region: (i) an UL data region, (ii) UL data region plus UL control region. [0104] In the DL control region, a PDCCH may be transmitted, and in the DL data region, a PDSCH may be transmitted. In the UL control region, a PUCCH may be transmitted, and in the UL data region, a PUSCH may be transmitted. In the PDCCH, Downlink Control Information (DCI), for example, DL data scheduling information or UL data scheduling information may be transmitted. In the PUCCH, Uplink Control Information (UCI), for example, ACK/NACK (Positive Acknowledgement/Negative Acknowledgement) information with respect to DL data, Channel State Information (CSI) information, or Scheduling Request (SR) may be transmitted. A GP provides a time gap during a process where a gNB and a UE transition from the transmission mode to the reception mode or a process where the gNB and UE transition from the reception mode to the transmission mode. Part of symbols belonging to the occasion in which the mode is changed from DL to UL within a subframe may be configured as the GP.

#### <Analog Beamforming #1>

[0105] Wavelengths are shortened in millimeter wave (mmW) and thus a large number of antenna elements can be installed in the same area. That is, the wavelength is 1 cm at 30 GHz and thus a total of 100 antenna elements can be installed in the form of a 2-dimensional array at an interval of 0.5 lambda (wavelength) in a panel of 5×5 cm. Accordingly, it is possible to increase a beamforming (BF) gain using a large number of antenna elements to increase coverage or improve throughput in mmW.

[0106] In this case, if a transceiver unit (TX RU) is provided to adjust transmission power and phase per antenna element, independent beamforming per frequency resource can be performed. However, installation of TX RUs for all of about 100 antenna elements decreases effectiveness in terms of cost. Accordingly, a method of mapping a large number of antenna elements to one TX RU and controlling a beam direction using an analog phase shifter is considered. Such analog beamforming can form only one beam direction in all bands and thus cannot provide frequency selective beamforming.

[0107] Hybrid beamforming (BF) having a number B of TX RUs which is smaller than Q antenna elements can be considered as an intermediate form of digital BF and analog BF. In this case, the number of directions of beams which can be simultaneously transmitted are limited to B although it depends on a method of connecting the B TXRUs and the Q antenna elements.

#### <Analog Beamforming #2>

[0108] When a plurality of antennas is used in NR, hybrid beamforming which is a combination of digital beamforming and analog beamforming is emerging. Here, in analog beamforming (or RF beamforming) an RF end performs precoding (or combining) and thus it is possible to achieve the performance similar to digital beamforming while reducing the number of RF chains and the number of D/A (or A/D) converters. For convenience, the hybrid beamforming structure may be represented by N TX RUs and M physical antennas. Then, the digital beamforming for the L data layers to be transmitted at the transmitting end may be represented by an N by L matrix, and the converted N digital signals are converted into analog signals via TX RUs, and analog beamforming represented by an M by N matrix is applied.

[0109] System information of the NR system may be transmitted in a broadcasting manner. In this case, in one symbol, analog beams belonging to different antenna panels may be simultaneously

transmitted. A scheme of introducing a beam RS (BRS) which is a reference signal (RS) transmitted by applying a single analog beam (corresponding to a specific antenna panel) is under discussion to measure a channel per analog beam. The BRS may be defined for a plurality of antenna ports, and each antenna port of the BRS may correspond to a single analog beam. In this case, unlike the BRS, a synchronization signal or an xPBCH may be transmitted by applying all analog beams within an analog beam group so as to be correctly received by any UE.

[0110] In the NR, in a time domain, a synchronization signal block (SSB, or also referred to as a synchronization signal and physical broadcast channel (SS/PB CH)) may consist of 4 OFDM symbols indexed from 0 to 3 in an ascending order within a synchronization signal block, and a primary synchronization signal (PSS), secondary synchronization signal (SSS), and a PBCH associated with demodulation reference signal (DMRS) may be mapped to the symbols. As described above, the synchronization signal block may also be represented by an SS/PBCH block.

[0111] In NR, since a plurality of synchronization signal blocks (SSBs) may be transmitted at different times, respectively, and the SSB may be used for performing initial access (IA), serving cell measurement, and the like, it is preferable to transmit the SSB first when transmission time and resources of the SSB overlap with those of other signals. To this purpose, the network may broadcast the transmission time and resource information of the SSB or indicate them through UE-specific RRC signaling.

[0112] In NR, transmission and reception may be performed based on beams. If reception performance of a current serving beam is degraded, a process of searching for a new beam through the so-called Beam Failure Recovery (BFR) may be performed.

[0113] Since the BFR process is not intended for declaring an error or failure of a link between the network and a UE, it may be assumed that a connection to the current serving cell is retained even if the BFR process is performed. During the BFR process, measurement of different beams (which may be expressed in terms of CSI-RS port or Synchronization Signal Block (SSB) index) configured by the network may be performed, and the best beam for the corresponding UE may be selected. The UE may perform the BFR process in a way that it performs an RACH process associated with a beam yielding a good measurement result.

[0114] Now, a transmission configuration indicator (hereinafter, TCI) state will be described. The TCI state may be configured for each CORESET of a control channel, and may determine a parameter for determining an RX beam of the UE, based on the TCI state.

[0115] For each DL BWP of a serving cell, a UE may be configured for three or fewer CORESETs. Also, a UE may receive the following information for each CORESET. [0116] 1) CORESET index  $p$  (e.g., one of 0 to 11, where index of each CORESET may be determined uniquely among BWPs of one serving cell), [0117] 2) PDCCH DM-RS scrambling sequence initialization value, [0118] 3) Duration of a CORESET in the time domain (which may be given in symbol units), [0119] 4) Resource block set, [0120] 5) CCE-to-REG mapping parameter, [0121] 6) Antenna port quasi co-location indicating quasi co-location (QCL) information of a DM-RS antenna port for receiving a PDCCH in each CORESET (from a set of antenna port quasi co-locations provided by a higher layer parameter called 'TCI-State'), [0122] 7) Indication of presence of Transmission Configuration Indication (TCI) field for a specific DCI format transmitted by the PDCCH in the CORESET, and so on.

[0123] QCL will be described. If a characteristic of a channel through which a symbol on one antenna port is conveyed can be inferred from a characteristic of a channel through which a symbol on the other antenna port is conveyed, the two antenna ports are said to be quasi co-located (QCLed). For example, when two signals A and B are transmitted from the same transmission antenna array to which the same/similar spatial filter is applied, the two signals may go through the same/similar channel state. From a perspective of a receiver, upon receiving one of the two signals, another signal may be detected by using a channel characteristic of the received signal.

[0124] In this sense, when it is said that the signals A and B are quasi co-located (QCLed), it may

mean that the signals A and B have went through a similar channel condition, and thus channel information estimated to detect the signal A is also useful to detect the signal B. Herein, the channel condition may be defined according to, for example, a Doppler shift, a Doppler spread, an average delay, a delay spread, a spatial reception parameter, or the like.

[0125] A 'TCI-State' parameter associates one or two downlink reference signals to corresponding QCL types (QCL types A, B, C, and D, see Table 4).

[Table 4]

TABLE-US-00005 TABLE 4 QCL Type Description QCL-TypeA Doppler shift, Doppler spread, Average delay, Delay spread QCL-TypeB Doppler shift, Doppler spread QCL-TypeC Doppler shift, Average delay QCL-TypeD Spatial Rx parameter

[0126] Each 'TCI-State' may include a parameter for configuring a QCL relation between one or two downlink reference signals and a DM-RS port of a PDSCH (or PDDCH) or a CSI-RS port of a CSI-RS resource.

[0127] Meanwhile, for each DL BWP configured to a UE in one serving cell, the UE may be provided with 10 (or less) search space sets. For each search space set, the UE may be provided with at least one of the following information.

[0128] 1) search space set index  $s$  ( $0 \leq s < 40$ ), 2) an association between a CORESET  $p$  and the search space set  $s$ , 3) a PDCCH monitoring periodicity and a PDCCH monitoring offset (slot unit), 4) a PDCCH monitoring pattern within a slot (e.g., indicating a first symbol of a CORSET in a slot for PDCCH monitoring), 5) the number of slots in which the search space set  $s$  exists, 6) the number of PDCCH candidates per CCE aggregation level, 7) information indicating whether the search space set  $s$  is CSS or USS.

[0129] In the NR, a CORESET #0 may be configured by a PBCH (or a UE-dedicated signaling for handover or a PSCell configuration or a BWP configuration). A search space (SS) set #0 configured by the PBCH may have monitoring offsets (e.g., a slot offset, a symbol offset) different for each associated SSB. This may be required to minimize a search space occasion to be monitored by the UE. Alternatively, this may be required to provide a beam sweeping control/data region capable of performing control/data transmission based on each beam so that communication with the UE is persistently performed in a situation where a best beam of the UE changes dynamically.

[0130] Meanwhile, in a wireless communication system, a UE receives information from a BS through a downlink (DL), and the UE transmits information to the BS through an uplink (UL). The information transmitted/received by the BS and the UE includes data and a variety of control information, and there are various physical channels according to a type/purpose of the information transmitted/received by the BS and the UE.

[0131] The UE which is powered on again in a power-off state or which newly enters a cell performs an initial cell search operation such as adjusting synchronization with the BS or the like (S11). To this end, the UE receives a primary synchronization channel (PSCH) and a secondary synchronization channel (SSCH) from the BS to adjust synchronization with the BS, and acquire information such as a cell identity (ID) or the like. In addition, the UE may receive a physical broadcast channel (PBCH) from the BS to acquire broadcasting information in the cell. In addition, the UE may receive a downlink reference signal (DL RS) in an initial cell search step to identify a downlink channel state.

[0132] Upon completing the initial cell search, the UE may receive a physical downlink control channel (PDCCH) and a physical downlink control channel (PDSCH) corresponding thereto to acquire more specific system information.

[0133] Thereafter, the UE may perform a random access procedure to complete an access to the BS. Specifically, the UE may transmit a preamble through a physical random access channel (PRACH), and may receive a random access response (RAR) for the preamble through a PDCCH and a PDSCH corresponding thereto. Thereafter, the UE may transmit a physical uplink shared channel (PUSCH) by using scheduling information in the RAR, and may perform a contention

resolution procedure similarly to the PDCCH and the PDSCH corresponding thereto.

[0134] After performing the aforementioned procedure, the UE may perform PDCCH/PDSCH reception and PUSCH/physical uplink control channel (PUCCH) transmission as a typical uplink/downlink signal transmission procedure. Control information transmitted by the UE to the BS is referred to as uplink control information (UCI). The UCI includes hybrid automatic repeat and request (HARQ) acknowledgement (ACK)/negative-ACK (NACK), scheduling request (SR), channel state information (CSI), or the like. The CSI includes a channel quality indicator (CQI), a precoding matrix indicator (PMI), a rank indication (RI), or the like. In general, the UCI is transmitted through the PUCCH. However, when control information and data are to be transmitted simultaneously, the UCI may be transmitted through the PUSCH. In addition, the UE may aperiodically transmit the UCI through the PUSCH according to a request/instruction of a network.

[0135] FIG. 10 shows an example of a downlink transmission/reception operation.

[0136] Referring to FIG. 10, the base station schedules downlink transmission such as frequency/time resources, transmission layer, downlink precoder, MCS, and the like (S1401). In particular, the base station may determine a beam for PDSCH transmission to the UE through the beam management operations described above. And, the UE receives downlink control information (DCI) for downlink scheduling (i.e., including scheduling information for the PDSCH) on the PDCCH from the base station (S1402). DCI format 1\_0 or 1\_1 may be used for downlink scheduling, and in particular, DCI format 1\_1 may include the following information: DCI format identifier (Identifier for DCI formats), bandwidth part indicator, frequency domain resource assignment, time domain resource assignment, PRB bundling size indicator, rate matching indicator, ZP CSI-RS trigger, antenna port(s), transmission configuration indication (TCI), SRS request, DMRS (Demodulation Reference Signal) sequence initialization.

[0137] In particular, according to each state indicated in the antenna port(s) field, the number of DMRS ports may be scheduled, in addition, single-user (SU)/multi-user (M U) transmission scheduling is possible. In addition, the TCI field consists of 3 bits, and the QCL for the DMRS is dynamically indicated by indicating a maximum of 8 TCI states according to the TCI field value. And, the UE receives downlink data from the base station on the PDSCH (S1403). If the UE detects a PDCCH including DCI format 1\_0 or 1\_1, it decodes the PDSCH according to an indication by the corresponding DCI.

[0138] Here, when the UE receives a PDSCH scheduled by DCI format 1, the UE may be configured with a DMRS configuration type by a higher layer parameter 'dmrs-Type', and the DMRS type is used to receive the PDSCH. In addition, the UE may be configured with the maximum number of front-loaded DMRS symbols for the PDSCH by the higher layer parameter 'maxLength'.

[0139] In the case of DMRS configuration type 1, when a single codeword is scheduled to the UE and an antenna port mapped with an index of {2, 9, 10, 11 or 30} is specified, or two codewords are scheduled to the UE, the UE assumes that all remaining orthogonal antenna ports are not associated with PDSCH transmission to another UE. Or, in the case of DMRS configuration type 2, when a single codeword is scheduled to the UE and an antenna port mapped with an index of {2, 10 or 23} is specified, or two codewords are scheduled to the UE, the UE assumes that all remaining orthogonal antenna ports are not associated with PDSCH transmission to another UE.

[0140] When the UE receives the PDSCH, it may be assumed that the precoding granularity  $P'$  is a consecutive resource block in the frequency domain. Here,  $P'$  may correspond to one of {2, 4, broadband}. If  $P'$  is determined to be broadband, the UE does not expect to be scheduled with non-contiguous PRBs, and the UE may assume that the same precoding is applied to the allocated resource. On the other hand, when  $P'$  is determined to be any one of {2, 4}, a precoding resource block group (PRG) is divided into  $P'$  consecutive PRBs. The actual number of consecutive PRBs in each PRG may be one or more. The UE may assume that the same precoding is applied to consecutive downlink PRBs in the PRG.

[0141] In order to determine a modulation order, a target code rate, and a transport block size in the PDSCH, the UE first reads the 5-bit MCD field in the DCI, and determines the modulation order and the target code rate. And, the UE reads the redundancy version field in the DCI, and determines the redundancy version. And, the UE determines the transport block size by using the number of layers and the total number of allocated PRBs before rate matching.

[0142] FIG. 11 shows an example of an uplink transmission/reception operation.

[0143] Referring to FIG. 11, the base station schedules uplink transmission such as frequency/time resources, transmission layer, uplink precoder, MCS, and the like (S1501). In particular, the base station can determine the beam for the UE for PUSCH transmission through the beam management operations described above. And, the UE receives the DCI for uplink scheduling (i.e., including the scheduling information of the PUSCH) on the PDCCH from the base station (S1502). DCI format 0\_0 or 0\_1 may be used for uplink scheduling, and in particular, DCI format 0\_1 may include the following information: DCI format identifier (Identifier for DCI formats), UL/SUL (Supplementary uplink) indicator (UL/SUL indicator), bandwidth part indicator, frequency domain resource assignment, time domain resource assignment, frequency hopping flag, modulation and coding scheme (MCS), SRS resource indicator (SRI), precoding information and number of layers, antenna port(s), SRS request, DMRS sequence initialization, UL-SCH (Uplink Shared Channel) indicator (UL-SCH indicator).

[0144] In particular, SRS resources configured in the SRS resource set associated with the higher layer parameter 'usage' may be indicated by the SRS resource indicator field. In addition, 'spatialRelationInfo' may be set for each SRS resource, and the value may be one of {CRI, SSB, SRI}.

[0145] Then, the UE transmits uplink data to the base station on PUSCH (S1503). When the UE detects a PDCCH including DCI format 0\_0 or 0\_1, it transmits a corresponding PUSCH according to an indication by the corresponding DCI. Two transmission schemes may be supported for PUSCH transmission, such as codebook-based transmission and non-codebook-based transmission.

[0146] In the case of codebook-based transmission, when the higher layer parameter 'txConfig' is set to 'codebook', the UE is set to codebook-based transmission. On the other hand, when the higher layer parameter 'txConfig' is set to 'nonCodebook', the UE is configured for non-codebook based transmission. If the higher layer parameter 'txConfig' is not set, the UE does not expect to be scheduled by DCI format 0\_1. When PUSCH is scheduled by DCI format 0\_0, PUSCH transmission is based on a single antenna port. In the case of codebook-based transmission, the PUSCH may be scheduled by DCI format 0\_0, DCI format 0\_1, or semi-statically scheduled. When this PUSCH is scheduled by DCI format 0\_1, as given by the SRS resource indicator field, and the precoding information and number of layers field, a PUSCH transmission precoder is determined based on SRI, TPMI (Transmit Precoding Matrix Indicator) and transmission rank from DCI. The TPMI is used to indicate a precoder to be applied across an antenna port, and corresponds to the SRS resource selected by the SRI when multiple SRS resources are configured. Alternatively, when a single SRS resource is configured, the TPMI is used to indicate a precoder to be applied across an antenna port, and corresponds to the single SRS resource. A transmission precoder is selected from the uplink codebook having the same number of antenna ports as the higher layer parameter 'nrofSRS-Ports'. When the higher layer set to 'codebook' is set by the parameter 'txConfig', the UE is configured with at least one SRS resource. The SRI indicated in slot n is associated with the most recent transmission of the SRS resource identified by the SRI, where the SRS resource precedes the PDCCH carrying the SRI (i.e., slot n).

[0147] For non-codebook-based transmission, the PUSCH may be scheduled by DCI format 0\_0, DCI format 0\_1, or semi-statically scheduled. When multiple SRS resources are configured, the UE may determine a PUSCH precoder and a transmission rank based on the wideband SRI. Here, the SRI is given by an SRS resource indicator in DCI or a higher layer parameter 'srs-ResourceIndicator'. The UE uses one or multiple SRS resources for SRS transmission. Here, the

number of SRS resources may be configured for simultaneous transmission in the same RB based on the UE capability. Only one SRS port is configured for each SRS resource. Only one SRS resource may be configured by the higher layer parameter ‘usage’ set to ‘nonCodebook’. The maximum number of SRS resources that can be configured for non-codebook-based uplink transmission is 4. The SRI indicated in slot  $n$  is associated with the most recent transmission of the SRS resource identified by the SRI, where the SRS transmission precedes the PDCCH carrying the SRI (i.e., slot  $n$ ).

[0148] In the case of NR, an uplink grant can be divided into (1) a dynamic grant (or with grant) and (2) a configured grant (or grant free or without grant).

[0149] FIG. 12 shows an example of an uplink grant.

[0150] Specifically, FIG. 12(a) shows an example of a dynamic grant, and FIG. 12(b) shows an example of a configured grant.

[0151] A dynamic grant refers to a scheduling-based data transmission/reception method of a base station in order to maximize resource utilization. This means that when there is data to be transmitted, the UE first requests uplink resource allocation to the base station, and can transmit data using only uplink resources allocated from the base station. In order to efficiently use uplink radio resources, the base station needs to know what type and how much of data to transmit in uplink for each UE. Accordingly, the UE directly transmits information about uplink data to transmit to the base station, and the base station can allocate uplink resources to the corresponding UE based on the information. In this case, the information on the uplink data transmitted from the UE to the base station is the amount of uplink data stored in its buffer, and is referred to as a buffer status report (BSR). The BSR is transmitted using a MAC control element when the UE is allocated a resource on the PUSCH in the current TTI and a reporting event is triggered.

[0152] Referring to FIG. 12(a), when an uplink radio resource for buffer status reporting (BSR) is not allocated to a UE, an uplink resource allocation process for actual data is exemplified. That is, in the case of a UE changing the state from the DRX mode to the active mode, since there is no pre-allocated data resource, resources for uplink data must be requested starting with SR transmission through PUCCH, and in this case, the uplink resource allocation procedure of 5 steps is used.

[0153] Referring to FIG. 12(a), in the case where the UE is not allocated a PUSCH resource for transmitting the BSR, the UE first transmits a scheduling request (SR) to the eNB in order to be allocated the PUSCH resource. When a reporting event occurs but the UE has not been scheduled radio resources on PUSCH in the current TTI, a scheduling request is used to request the UE to receive PUSCH resources for uplink transmission. That is, when a regular buffer status report (regular BSR) is triggered, but the UE does not have an uplink radio resource for transmitting the BSR to the base station, the SR is transmitted on the PUCCH. Depending on whether the PUCCH resource for the SR is configured, the UE transmits the SR through the PUCCH or initiates a random access procedure. Specifically, the PUCCH resource through which the SR can be transmitted is configured by a UE-specific higher layer (e.g., RRC layer), and the SR configuration includes an SR transmission period (SR periodicity) and an SR subframe offset information. Upon receiving the UL grant for the PUSCH resource for BSR transmission from the base station, the UE transmits the triggered BSR through the PUSCH resource allocated by the UL grant to the base station. The base station checks the amount of data that the UE actually transmits in uplink through the BSR, and transmits a UL grant for a PUSCH resource for actual data transmission to the UE. Upon receiving the UL grant for actual data transmission, the UE transmits actual uplink data to the base station through the allocated PUSCH resource.

[0154] Referring to FIG. 12(b), a configured grant method will be described.

[0155] The UE receives a resource configuration for transmission of UL data without a grant from the base station. The resource configuration may be performed only with RRC signaling (type 1), or may be performed with L1 (layer-1) signaling and RRC signaling (type 2). And, the UE

performs initial transmission to the base station based on the resource configuration received without the grant. In this case, the initial transmission may be repeated, and repetition of the initial transmission for the same transport block may be performed  $K$  times ( $K \geq 1$ ).

[0156] Resources for initial transmission by a configured grant may or may not be shared between one or more UEs.

[0157] When the initial transmission by the configured grant fails, the base station may transmit a grant for retransmission of a TB related to the initial transmission to the UE. At this time, the base station needs to identify the UE even if a collision occurs. A UE performing UL transmission without a UL grant may be identified based on time/frequency resources and reference signal (RS) parameters.

[0158] The base station may allocate different DMRS resources to different UEs sharing the same PUSCH resource. And, when the UE performs retransmission, it is switched to the grant-based transmission, the UE receives a grant from the base station, and performs retransmission based on the grant. That is, the UE performs initial transmission without a grant, but performs retransmission based on a grant.

[0159] FIG. 13 shows an example of a grant-free initial transmission.

[0160] Referring to FIG. 13, the initial transmission may be performed with a period of  $P$ . The initial transmission may be a grant-free transmission, and subsequent repeated transmissions may be a grant-based transmission.

[0161] In order to enable reasonable battery consumption when bandwidth adaptation (BA) is configured, only one uplink BWP and one downlink BWP or only one downlink/uplink BWP pair for each uplink carrier may be activated at once in an active serving cell, and all other BWPs configured in the UE are deactivated. In the deactivated BWPs, the UE does not monitor the PDCCH, and does not perform transmission on the PUCCH, PRACH, and UL-SCH.

[0162] For the BA, RX and TX bandwidths of the UE are not necessarily as wide as a bandwidth of a cell, and may be adjusted. That is, it may be commanded such that a width is changed (e.g., reduced for a period of low activity for power saving), a position in a frequency domain is moved (e.g., to increase scheduling flexibility), and a subcarrier spacing is changed (e.g., to allow different services). A subset of the entire cell bandwidth of a cell is referred to as a bandwidth part (BWP), and the BA is acquired by configuring BWP(s) to the UE and by notifying the UE about a currently active BWP among configured BWPs. When the BA is configured, the UE only needs to monitor the PDCCH on one active BWP. That is, there is no need to monitor the PDCCH on the entire downlink frequency of the cell. A BWP inactive timer (independent of the aforementioned DRX inactive timer) is used to switch an active BWP to a default BWP. That is, the timer restarts when PDCCH decoding is successful, and switching to the default BWP occurs when the timer expires.

[0163] Hereinafter, the present disclosure proposes a cross-slot scheduling method and an apparatus using the method in a wireless communication system.

[0164] In NR, a power saving technique is being discussed to reduce the power consumption of the UE, and among the techniques, there is a power saving technique using cross-slot scheduling.

[0165] A power saving technique using cross-slot scheduling indicates a minimum slot offset between the DCI and the PDSCH scheduled in the DCI. Power consumption can be reduced by a method in which the UE (micro-) sleeps or applies PDCCH decoding relaxation (e.g., using a low voltage/low clock speed) in a duration guaranteed by the minimum slot offset. For example, assume that the minimum slot offset is 2. In this case, if the UE receives DCI in slot  $\#N$ , the UE sleeps in slot  $\#N+1$  and wakes up in slot  $\#N+2$  to receive the PDSCH scheduled by the DCI. Since the minimum slot offset is 2, it is okay to sleep in slot  $\#N+1$ . The minimum slot offset may be referred to as a minimum applicable slot offset or a minimum applicable offset, a minimum scheduling offset, or the like.

[0166] As a specific example, the network may set a minimum applicable  $K=0/K=2$  value to indicate to the UE the minimum slot offset between the DCI and the corresponding scheduled

PDSCH/PUSCH when scheduling a PDSCH or a PUSCH. Here, K 0 may be an offset (slot offset) related to a time relationship between a slot in which DCI is received and a slot in which a PDSCH scheduled by the DCI is received. K 2 may be based on the numerology of the PDSCH. K 2 may be an offset (slot offset) related to a time relationship between a slot in which DCI is received and a slot in which a PUSCH scheduled by the DCI is transmitted. K 2 may be based on the numerology of PUSCH. It can be seen that the minimum applicable K 0 indicates the minimum applicable value (restriction) in setting the K 0 value, and the minimum applicable K 2 indicates the minimum applicable value (restriction) in the setting of the K 2 value. Hereinafter, the minimum applicable K 0 may be expressed as K 0 min, and the minimum applicable K 2 may be expressed as K 2 min.

[0167] For example, the base station may instruct K 0 and K 2 to the UE in the following manner.

[0168] When the UE is scheduled to receive the PDSCH by DCI, the time domain resource assignment field value m of DCI provides the row index m+1 in the resource allocation table. The indexed row directly defines the slot offset K 0, the start and length indicator SLIV or the start symbol S and the allocation length L, and the PDSCH mapping type assumed in PDSCH reception.

[0169] The following table is an example of a resource allocation table.

[Table 5]

TABLE-US-00006 TABLE 5 dmrs-TypeA- PDSCH Row index Position mapping type K.sub.0 S L  
1 2 Type A 0 2 12 3 Type A 0 3 11 2 2 Type A 0 2 10 3 Type A 0 3 9 3 2 Type A 0 2 9 3 Type A 0 3 8  
4 2 Type A 0 2 7 3 Type A 0 3 6 5 2 Type A 0 2 5 3 Type A 0 3 4 6 2 Type B 0 9 4 3 Type B 0 10 4 7  
2 Type B 0 4 4 3 Type B 0 6 4 8 2, 3 Type B 0 5 7 9 2, 3 Type B 0 5 2 10 2, 3 Type B 0 9 2 11 2, 3  
Type B 0 12 2 12 2, 3 Type A 0 1 13 13 2, 3 Type A 0 1 6 14 2, 3 Type A 0 2 4 15 2, 3 Type B 0 4 7  
16 2, 3 Type B 0 8 4

[0170] Given the parameter value of the indexed row, the slot allocated to the PDSCH is  $\text{floor}(n \cdot \text{Math}((2 \cdot \text{sup} \cdot \mu \text{PDSCH} / 2 \cdot \text{sup} \cdot \mu \text{PDCCH})) + K 0)$ . Here, n is a slot with a scheduling DCI, and K 0 is based on the numerology of the PDSCH. Each of  $\mu \text{P.sub.DSCH}$  and  $\mu \text{P.sub.DCCH}$  is a subcarrier spacing configuration for each of PDSCH and PDCCH.

[0171] The start symbol S for the start of the slot and L (the number of symbols allocated to the PDSCH), which is the number of consecutive symbols counted from the symbol S, are determined from the 'start and length indicator' SLIV.

[0172] When the UE is scheduled to transmit a transport block but no CSI report, or when the UE is scheduled to transmit a transport block and CSI report(s) in PUSCH by DCI, the time domain resource assignment field value m of the DCI provides the row index m+1 in the allocated table. The indexed row directly defines a slot offset K 2, a start and length indicator SLIV or a start symbol S and an allocation length L, and a PUSCH mapping type to be applied to PUSCH transmission.

[0173] The slot in which the UE should transmit the PUSCH may be determined, by K 2, as  $\text{floor}(n \cdot \text{Math}((2 \cdot \text{sup} \cdot \mu \text{PUSCH} / 2 \cdot \text{sup} \cdot \mu \text{PDCCH})) + K 2)$ . Here, n is a slot with a scheduling DCI, and K 2 is based on the numerology of PUSCH.  $\mu \text{P.sub.PUSCH}$  and  $\mu \text{P.sub.DCCH}$  are subcarrier spacing configurations for PUSCH and PDCCH, respectively.

[0174] The start symbol S for the start of a slot (relative to the start of the slot) and L, the number of consecutive symbols allocated to the PUSCH counted from the symbol S, is determined from the start and length indicator SLIV of the indexed row.

[0175] On the other hand, in the case of instructing/changing the minimum applicable K 0/K 2 (K 0 min/K 2 min), when the corresponding instruction/change is applied may be determined by "application delay", and the application delay may be defined as follows. Hereinafter, for convenience, the application delay may be expressed as X or the application delay X.

[0176] For the application delay X to apply the minimum applicable K 0/K 2 value(s) indicated for the scheduled cell, triggered by a 1-bit indication of DCI format 1\_1 or 0\_1 in the scheduling cell,

[0177] the UE receives DCI indicating a change of the minimum applicable K 0/K 2 value in slot n of the scheduling cell, [0178] the UE may assume that a new minimum applicable K 0/K 2 value is

applied to the PDSCH/PUSCH of the scheduled cell from the slot ( $n+X$ ) of the scheduling cell. That is, when DCI indicating a change of the minimum applicable  $K_0/K_2$  value is received in slot  $n$  of the scheduling cell, the changed minimum applicable  $K_0/K_2$  value is applied from slot  $n+X$  of the scheduling cell.

[0179] In case of self-carrier scheduling and at least PDCCH monitoring case 1-1 (to be described later),  $X=\max(Y, Z)$ . Here,  $Y$  is the active minimum applicable  $K_0$  value of the active DL BWP before the change indication,  $Z$  is (1, 1, 2, 2) for each downlink subcarrier spacing (DL SCS) (15, 30, 60, 120) kHz, respectively.

[0180] In the above definition,  $Z$  may be defined as “the minimum feasible non-zero application delay that may depend on DL SCS”, and it might be interpreted as the minimum time for PDCCH decoding.

[0181] The  $Z$  value is applied when  $Y$  is 0 or less than  $Z$ . In this case, the new minimum applicable value  $K_0/K_2$  ( $K_0 \text{ min}/K_2 \text{ min}$ ) may mean that it is applied from the point in time when the UE recognizes the corresponding value (i.e., the time when PDCCH decoding ends).

[0182] FIG. 14 is an example of applying an application delay.

[0183] Referring to FIG. 14(a), the UE receives DCI in slot  $m$ , and receives the PDSCH scheduled by the DCI in slot  $m+K_0$ . In this case, for example, it is assumed that the  $K_0 \text{ min}$  value that is the minimum applicable slot offset is 1 and the  $K_0$  value is 2. Then, the UE may receive DCI including information indicating a change of the minimum applicable slot offset in slot  $n$ . Through this, for example, suppose that the value of  $K_0 \text{ min}$  is changed to 0. In this case, the change of the minimum applicable slot offset is applied from slot  $n+X$ , not from slot  $n$ . The value of  $X$  may be determined to be a largest value among  $Y$  and  $Z$ , such as  $\max(Y, Z)$ . Here,  $Y$  is the active minimum applicable  $K_0$  value of the active DL BWP before the change indication,  $Z$  may be (1, 1, 2, 2) in turn for the case where the DL SCS is (15, 30, 60, 120) kHz. In slot  $n+X$ , the UE may receive a DCI indicating 0 as a value of  $K_0$ , and may also receive a PDSCH scheduled by the DCI.

[0184] Referring to FIG. 14(b), the UE may receive DCI #1 including information indicating a change of the minimum applicable slot offset in slot  $n$ . In this case, the changed minimum applicable slot offset (let's call it  $K_0 \text{ minNew}$ ) is applied from slot  $n+X$ , not directly applied from slot  $n$ . That is, DCI #2 received in slot  $n+X$  indicates a  $K_0$  value greater than or equal to  $K_0 \text{ minNew}$ . From the UE's point of view, from slot  $n+X$ , it is expected to receive a DCI indicating a  $K_0$  value greater than or equal to  $K_0 \text{ minNew}$ . Until slot  $n+X$ , the existing minimum applicable slot offset (let's call this  $K_0 \text{ minOld}$ ) is applied.

[0185] PDCCH monitoring case 1-1, case 1-2, and case 2 may be defined as follows.

[0186] Case 1: a case that the PDCCH monitoring period is 14 or more symbols.

[0187] Case 1-1: a case that PDCCH monitoring is performed on up to 3 OFDM symbols from the beginning of a slot.

[0188] Case 1-2: a case that PDCCH monitoring is performed on any up to 3 consecutive OFDM symbols of a slot.

[0189] For a given UE, all search space configurations are within the same range of three consecutive OFDM symbols in a slot.

[0190] Case 2: a case that the PDCCH monitoring period is less than 14 symbols. This includes monitoring the PDCCH on up to 3 OFDM symbols from the beginning of the slot.

[0191] This disclosure proposes a method of defining ‘application delay’ in each case and cross-carrier scheduling.

<Application Delay>

[0192] FIG. 15 illustrates a location of a CORESET for PDCCH monitoring.

[0193] Referring to FIG. 15, the first CORESET 151 for PDCCH monitoring is located within the first three symbols of the slot, and the second CORESET 152 for PDCCH monitoring is located outside the first three symbols of the slot, for example, may be located in the last three symbols of the slot. It can be seen that the first CORESET 151 corresponds to the aforementioned cases 1-1

and 1-2, and the second CORESET 152 corresponds to the aforementioned cases 1-2.

[0194] That is, in case 1-2, unlike case 1-1 (the CORESET for PDCCH monitoring is located within the first 3 symbols in the slot), there is no position restriction in the slot of the CORESET for monitoring the PDCCH. As shown in FIG. 15, the network may instruct the UE to position the CORESET in the last 3 symbols in the slot and monitor the PDCCH. This means that the PDCCH decoding end time (i.e., DCI decoding end time) may be different depending on the location of the CORESET. In determining the application delay X value for indicating the change of the minimum applicable value  $K_0/K_2$  ( $K_0$  min/ $K_2$  min) through DCI and determining the application timing, this point should be considered.

[0195] For example, when  $Y=0$ ,  $Z=1$ ,  $X=\max(Y, Z)=1$  is given. This means that the changed minimum applicable value  $K_0/K_2$  ( $K_0$  min/ $K_2$  min) is applied to the slot immediately following the change indicated. However, there may be cases where this is not possible.

[0196] For example, PDCCH decoding may end in the next slot. For example, if the CORESET for PDCCH monitoring is located in the last 3 symbols of a slot, the UE will receive the PDCCH in the last 3 symbols and then decode the PDCCH in the next slot. Therefore, it may not be possible to apply the PDCCH decoding result from the beginning of the next slot.

[0197] In order to solve such a problem, the present disclosure proposes the following method. A solution method using the Z value is proposed below, but the same method may be applied to the X or Y value.

[0198] Option 1) It can be applied by adding a specific value (e.g., 1) to the predefined Z value.

[0199] Option 1 is the simplest solution, and in cases 1-2, the application delay can be derived by adding a specific value (e.g., 1) to the Z value. In this case, the specific value may be predefined or indicated through higher layer signaling (e.g., RRC, MAC CE, etc.) of the network.

[0200] Option 2) Determining the Z value based on the location of the CORESET (group).

[0201] The UE may determine whether to add a specific value (e.g., 1) to the Z value according to the location of the CORESET to be monitored in the corresponding slot. To this end, the location of the CORESET serving as a reference may be defined in advance or may be indicated by higher layer signaling of the network. Alternatively, the reference position of the CORESET may be determined according to the decoding capability of the UE. In this case, the UE may report the decoding capability (e.g., the position of the CORESET capable of terminating PDCCH decoding within the corresponding slot).

[0202] As the CORESET, all CORESETs monitored in the corresponding slot may be considered, or the CORESET, where non-fallback DCI to which the minimum applicable  $K_0/K_2$  may be indicated is monitored, may be limitedly applied to.

[0203] For example, when all or part of the CORESET (in which PDCCH monitoring needs to be performed) exists after a specific symbol index (indicated in advance, or indicated by higher layer signaling of the network, or indicated by the capability reported by the UE), the UE may derive an application delay by adding a specific value (e.g., 1) to a predefined Z value.

[0204] Option 3) Z values specific to the case 1-2.

[0205] The network may separately indicate the Z value to be applied to Case 1-2. Alternatively, the Z value for Case 1-2 may be determined by a predefined definition. In addition, as in option 2, it may determine whether to apply the Z value for the case 1-2 depending on the location of the CORESET, in option 3.

Case 2.

[0206] Case 2 refers to a case in which a plurality of monitoring occasions of a specific search space set are set (or can be set) in one slot. In case 2, the following method can be considered.

[0207] Option 1) A method that does not apply the power saving technique using cross-slot scheduling to case 2.

[0208] As described above, the power saving technique using cross-slot scheduling is a scheme of performing a power saving operation during a slot offset between the PDCCH and the scheduled

PDSCH. However, in case 2, since one search space set may have a plurality of monitoring occasions within one slot, it may be difficult to expect power savings due to sleep or the like. Therefore, in case 2, it can be assumed that the power saving operation by the minimum applicable  $K_0$  is not performed.

[0209] In addition, the PDCCH monitoring occasion is determined by the search space set setting. When a plurality of search space sets are configured, different cases may be applied to each slot. Therefore, when different cases are applied to each slot, option 1 may be interpreted as suggesting that it is assumed that cross-slot scheduling is not applied in the slot corresponding to case 2 or that the minimum applicable value  $K_0/K_2$  is not changed in the slot corresponding to case 2.

[0210] As another method, when the minimum applicable value  $K_0/K_2$  is newly indicated (i.e., changed) by DCI transmitted in the slot corresponding to case 2, the UE may ignore the indication. This method can be applied to case 1-2 as well as case 2.

[0211] Option 2) A method of applying an application delay for each monitoring occasion

[0212] Since the application delay means the time when the newly indicated minimum applicable values  $K_0/K_2$  are applied, a method of defining the application delay for each monitoring occasion may also be considered. Therefore, in case 2, application delay can be applied for each monitoring occasion, this may mean that the application delay derivation method applied to Case 1-1 and Case 1-2 is applied according to the location of the monitoring occasion in the slot.

<Cross Carrier Scheduling>

[0213] Cross-carrier scheduling refers to a method of scheduling a PDSCH of a scheduled cell (a cell which is scheduled) in a PDCCH of a scheduling cell (a cell performing scheduling). That is, the PDCCH for PDSCH scheduling is decoded in the scheduling cell (more specifically, the active DL BWP of the scheduling cell), and scheduling of the PDSCH transmitted in the scheduled cell (more specifically, the active DL BWP of the scheduled cell) is performed through the corresponding DCI.

[0214] When applying a power saving technique using cross-slot scheduling in a scheduled cell (active BWP), the application delay for the minimum applicable value  $K_0/K_2$  of the scheduled cell (active BWP) may be defined in the following method.

Option 1) Scheduling Cell (Active BWP) Based Application Delay

[0215] The application delay can be interpreted as an offset from the slot in which the DCI indicating the new minimum applicable  $K_0/K_2$  is transmitted to the slot to which the corresponding value is actually applied. This is closely related to PDCCH decoding. As described above, since the cross-carrier scheduling is a process of scheduling the PDSCH of a scheduled cell through the PDCCH in the scheduling cell, it may be desirable to replace the application delay of the minimum applicable value  $K_0/K_2$  of the scheduled cell with the application delay of the scheduling cell (active BWP). Therefore, the present disclosure proposes to determine Y and/or Z, which are parameters for determining the application delay, based on the scheduling cell.

[0216] For example, in cross-carrier scheduling, the application delay of the scheduled cell may be determined as follows. (Alternatively, it may be defined to follow the application delay of the scheduling cell.)

[0217] For the application delay X to apply the minimum applicable  $K_0/K_2$  value(s) indicated for the scheduled cell, triggered by a 1-bit indication of DCI format 1\_1 or 0\_1 in the scheduling cell, [0218] the UE receives DCI indicating a change in slot n of the scheduling cell, [0219] the UE may be scheduled with the minimum applicable  $K_0/K_2$  value for the PDSCH/PUSCH of the scheduled cell in DCI in the slot (n+X) of the scheduling cell.

[0220] For cross-carrier scheduling,  $X = \max(Y, Z)$ . Here, Y is the active minimum applicable  $K_0$  value of the active DL BWP of the scheduling cell, and Z is (1, 1, 2, 2) for each downlink subcarrier spacing (DL SCS) (15, 30, 60, 120) kHz of the active BWP of the scheduling cell, respectively.

Option 2) Application Delay Based on Scheduled Cell (Active BWP)

[0221] When the UE separately performs processing (e.g., PDCCH decoding) for the scheduling cell and the scheduled cell, the PDCCH scheduling the PDSCH of the scheduled cell is transmitted in the scheduling cell, but the application delay for the change of the minimum applicable value of the scheduled cell may be determined based on the scheduled cell. However, in this case, if the scheduling cell and the scheduled cell have different numerology, a process of scaling to fit the numerology of the scheduling cell may be required. For example, in cross-carrier scheduling, the application delay of the scheduled cell may be determined as follows.

[0222] In the application delay  $X$  for applying the minimum applicable  $K_0/K_2$  value(s) indicated for the scheduled cell triggered by the 1-bit indication of DCI format 1\_1 or 0\_1 in the scheduling cell, [0223] the UE receives DCI indicating a change in slot  $n$  of the scheduling cell, [0224] the UE may be scheduled with the minimum applicable  $K_0/K_2$  value for the PDSCH/PUSCH of the scheduled cell in DCI in the slot  $(n+X)$  of the scheduling cell.

[0225] For cross-carrier scheduling,  $X = \max(Y, Z) \cdot \text{Math.}(2^{\mu_{\text{sub.scheduling}}}/2^{\mu_{\text{sub.scheduled}}})$  or  $X = \text{ceil}(\max(Y, Z) \cdot (2^{\mu_{\text{sub.scheduling}}}/2^{\mu_{\text{sub.scheduled}}}))$ . Here,  $Y$  is the active minimum applicable  $K_0$  value of the active DL BWP of the scheduled cell before the change indication. For each downlink subcarrier spacing (DL SCS) (15, 30, 60, 120) kHz of the active BWP of the scheduled cell,  $z$  is (1, 1, 2, 2) respectively.

[0226] In the above formula,  $\mu_{\text{sub.scheduling}}$  denotes the scheduling cell's numerology (subcarrier spacing configuration), and  $\mu_{\text{sub.scheduled}}$  denotes the scheduled cell's numerology (subcarrier spacing configuration). For {15 kHz, 30 kHz, 60 kHz, 120 kHz}, it may have values of {0, 1, 2, 3} respectively.

### Option 3) Combination of Option 1 and 2

[0227] Parameters  $Y$  and  $Z$  may be determined based on a scheduled cell and a scheduling cell, respectively. For example, since  $Y$  means the minimum applicable value  $K_0/K_2$  before change, it is determined based on the scheduled cell to which the minimum applicable value is applied. And  $Z$  may be determined based on a scheduling cell in which actual PDCCH decoding is performed.

[0228] In cross-carrier scheduling, the application delay of the scheduled cell may be determined as follows.

[0229] In the application delay  $X$  for applying the minimum applicable  $K_0/K_2$  value(s) ( $K_0 \text{ min}/K_2 \text{ min}$ ) indicated for the scheduled cell triggered by the 1-bit indication of DCI format 1\_1 or 0\_1 in the scheduling cell, [0230] the UE receives DCI indicating a change in slot  $n$  of the scheduling cell, [0231] the UE may be scheduled with the minimum applicable  $K_0/K_2$  value for the PDSCH/PUSCH of the scheduled cell in DCI in the slot  $(n+X)$  of the scheduling cell.

[0232] For cross-carrier scheduling,  $X = \max(Y \cdot \text{Math.}(2^{\mu_{\text{sub.scheduling}}}/2^{\mu_{\text{sub.scheduled}}}), Z)$  or  $X = \max(\text{ceil}(Y \cdot \text{Math.}(2^{\mu_{\text{sub.scheduling}}}/2^{\mu_{\text{sub.scheduled}}}), Z)$ . Here,  $Y$  is the active minimum applicable  $K_0$  value of the active DL BWP of the scheduled cell before the change indication,  $Z$  is (1, 1, 2, 2) for each downlink subcarrier spacing (DL SCS) (15, 30, 60, 120) kHz of the active BWP of the scheduling cell, respectively.

[0233] FIG. 16 illustrates a method for determining an application delay value according to option 3.

[0234] Referring to FIG. 16, the UE receives downlink control information (DCI) including information indicating a change to the value of  $K_0 \text{ min}$  or  $K_2 \text{ min}$  in slot  $n$  of a scheduling cell (S161). For example, in slot  $n$  of a scheduling cell, the first DCI including information indicating a change to the value of  $K_2 \text{ min}$ , which is the minimum scheduling offset, may be received.

[0235] The DCI may be received in symbols before a specific symbol index of the slot  $n$  (e.g., the first 3 symbols of the slot  $n$ ).

[0236] Each of the  $K_0 \text{ min}$  and  $K_2 \text{ min}$  is an applied minimum scheduling offset restriction. Specifically, the  $K_0 \text{ min}$  may be a minimum scheduling offset restriction related to a minimum value of an offset between a slot for receiving the first DCI and a slot for receiving a physical downlink shared channel (PDSCH) scheduled by the first DCI, the  $K_2 \text{ min}$  may be a minimum

scheduling offset restriction related to a minimum value of an offset between a slot for receiving the second DCI and a slot for transmitting a physical uplink shared channel (PUSCH) scheduled by the second DCI.

[0237] The UE may determine the application delay X as a largest value among i) a first value obtained by multiplying currently applied K 0 min (let's call this Y) in a scheduled cell scheduled by the first DCI by  $2 \cdot \frac{\mu_{\text{scheduling}}}{\mu_{\text{scheduled}}}$  and then performing ceiling and ii) a second value (let's call this Z) that are predetermined depending on a subcarrier spacing (SCS) of the scheduling cell (S162).

[0238] That is, the application delay X can be determined by the following equation.

[00001]  $X = \max(\text{Math. } Y \cdot \text{Math. } \frac{2 \cdot \mu_{\text{scheduling}}}{\mu_{\text{scheduled}}}, Z)$  [Equation1]

[0239] The  $\mu_{\text{sub.scheduling}}$  is a subcarrier spacing configuration of the scheduling cell (That is, subcarrier spacing configuration associated with PDCCH, therefore,  $\mu_{\text{sub.scheduling}}$  may be expressed as UPDCCH) and the  $\mu_{\text{sub.scheduled}}$  is a subcarrier spacing configuration of the scheduled cell (That is, subcarrier spacing configuration associated with PDSCH, therefore,  $\mu_{\text{sub.scheduled}}$  may be expressed as  $\mu_{\text{P.sub.DSCH}}$ ). The Y is the K 0 min value currently applied to the scheduled cell, and the Z is the second value.

[0240] Z may be predetermined as shown in the following table depending on the subcarrier spacing (SCS) (or subcarrier spacing configuration u) of the scheduling cell.

TABLE-US-00007 TABLE 6  $\mu$  Z 0 1 1 2 2 3 2

[0241] That is, when the subcarrier spacing (SCS) of the scheduling cell is 15, 30, 60, and 120 kHz, the Z value may be predetermined as 1, 1, 2, 2, respectively.

[0242] The UE applies the changed K 0 min or the changed K 2 min value from after the slot n+X of the scheduling cell (S163). For example, the UE can expect to receive the second DCI based on the changed K 2 min value from after slot n+X of the scheduling cell. From the network point of view, from slot n+X of the scheduling cell, the second DCI based on the changed K 2 min value is transmitted.

[0243] Thereafter, the UE transmits the PUSCH scheduled by the second DCI. The offset (difference) between the slot in which the second DCI is received and the slot in which the PUSCH scheduled by the second DCI is transmitted should be equal to or greater than the changed K 2 min. The UE may perform a power saving operation between the slot in which the second DCI is received and the slot in which the PUSCH scheduled by the second DCI is transmitted.

[0244] On the other hand, as described above in 'Option 2) Determining the Z value based on the location of the CORESET (group)', if the DCI is received in symbols after the specific symbol index of slot n (e.g., symbols outside the first 3 symbols of slot n), after increasing the second value Z by 1, the value of X is determined. The reason for doing this is considering that the decoding (complete) timing of the DCI may be slot n+1 instead of slot n depending on the location of the CORESET.

[0245] For example, let assume that downlink control information (DCI) including information indicating a change in the value of K 0 min or K 2 min is received in the last three symbols of slot n. In this case, Equation 1 is used in obtaining the application delay X, but the Z value from Table 6 is incremented by one (i.e., Z+1), instead of the Z value from Table 6, and then used for Z of Equation 1.

[0246] FIG. 17 is an example of applying the method of FIG. 16.

[0247] Referring to FIG. 17, it is assumed that the subcarrier spacing (SCS) configuration of the scheduling cell is  $\mu=0$ , and the SCS configuration  $\mu=1$  of the scheduled cell. It is assumed that K 0 min (i.e., Y) currently applied to the scheduled cell is referred to as K 0 minOld for convenience, and its value is 1. Since the subcarrier spacing (SCS) configuration of the scheduling cell  $\mu=0$ , Z=1.

[0248] DCI including information indicating a change to K 0 min may be received within the first three symbols of the slot n of the active DL BWP of the scheduling cell. Also, assume that the DCI

is a DCI for cross-carrier scheduling.

[0249] The DCI may be, for example, DCI format 0\_1 for scheduling one or more PUSCHs and DCI format 1\_1 for scheduling PDSCH. Each of DCI format 0\_1 and DCI format 1\_1 may or may not include a 1-bit 'Minimum applicable scheduling offset indicator', FIG. 17 exemplifies an inclusion case. In case of DCI format 0\_1, when the value of 'Minimum applicable scheduling offset indicator' is 0, it indicates the first value among the  $K_{\text{sub.2min}}$  values set by the higher layer signal. When the value of 'Minimum applicable scheduling offset indicator' is 1, it indicates the second value (if any) or 0 (if there is no second value) among the  $K_{\text{2 min}}$  values set by the higher layer signal. In case of DCI format 1\_1, when the value of 'Minimum applicable scheduling offset indicator' is 0, it indicates the first value among the  $K_{\text{0 min}}$  values set by the higher layer signal. When the value of 'Minimum applicable scheduling offset indicator' is 1, it indicates the second value (if any) or 0 (if there is no second value) among the  $K_{\text{0 min}}$  values set by the higher layer signal.

[0250] It is possible to determine whether a change in the  $K_{\text{0 min}}$ / $K_{\text{2 min}}$  value is indicated according to the value of  $K_{\text{0 min}}$ / $K_{\text{2 min}}$  indicated by the value of 'Minimum applicable scheduling offset indicator'.

[0251] When the DCI indicates a change in the value of  $K_{\text{0 min}}$  ( $K_{\text{2 min}}$ ), the time point at which the changed  $K_{\text{0 min}}$  ( $K_{\text{2 min}}$ ) is applied is slot  $n+X$ , where  $X$  is  $X=\max(\text{ceil}(1.\text{Math.2.sup.0}/2.\text{sup.1}),1)=1$  according to Equation 1 above. Accordingly, the time point at which the changed  $K_{\text{0 min}}$ / $K_{\text{2 min}}$  is applied becomes slot  $n+1$ .

[0252] FIG. 18 is another example of applying the method of FIG. 16.

[0253] Referring to FIG. 18, it is assumed that the subcarrier spacing (SCS) configuration of the scheduling cell is  $\mu=2$ , and the SCS configuration of the scheduled cell is  $\mu=1$ . It is assumed that  $K_{\text{0 min}}$  (i.e.,  $Y$ ) currently applied to the scheduled cell is referred to as  $K_{\text{0 minOld}}$  for convenience, and its value is 1. Since the subcarrier spacing (SCS) configuration of the scheduling cell is  $\mu=2$ ,  $Z=2$ .

[0254] DCI including information indicating a change of  $K_{\text{0 min}}$  may be received within the first three symbols of the slot  $n$  of the active DL BWP of the scheduling cell. Also, assume that the DCI is a DCI for cross-carrier scheduling.

[0255] When the DCI indicates a change in the value of  $K_{\text{0 min}}$  ( $K_{\text{2 min}}$ ), the time point at which the changed  $K_{\text{0 min}}$  ( $K_{\text{2 min}}$ ) is applied is slot  $n+X$ , where  $X$  is, according to Equation 1,  $X=\max(\text{ceil}(1,2.\text{sup.2}/2.\text{sup.1}),2)=2$ . Accordingly, the time point at which the changed  $K_{\text{0 min}}$ / $K_{\text{2 min}}$  is applied becomes slot  $n+2$ .

[0256] FIG. 19 illustrates a signaling method between a network (base station) and a UE.

[0257] Referring to FIG. 19, the base station provides a higher layer signal for setting  $K_{\text{2 min}}$  values to the UE (S191). For example, 'minimumSchedulingOffsetK 2' may be provided through 'PUSCH-Config' used to set UE-specific PUSCH parameters, and 'minimumSchedulingOffsetK 2' may include a list of  $K_{\text{2 min}}$  values.

[0258] The base station transmits a first DCI including information indicating a change to  $K_{\text{2 min}}$  to the UE in slot  $n$  of the scheduling cell (S192). The first DCI may be DCI format 0\_1 or DCI format 1\_1. The first DCI may be transmitted within the first three symbols of slot  $n$  or outside the first three symbols, and the  $Z$  value used to determine the application delay  $X$  may vary depending on where it is transmitted. This has already been described above. The first DCI may inform the change of  $K_{\text{2 min}}$  through a 1-bit field. This has already been described above.

[0259] The UE determines the application delay value  $X$  related to the time to apply the changed  $K_{\text{2 min}}$  (S193). As described above, the  $X$  value may be determined based on the current  $K_{\text{0 min}}$  of the scheduled cell, the SCS configurations of the scheduling cell and the scheduled cell, and a predetermined value dependent on the SCS configuration of the scheduling cell. For example, Equation 1 can be used.

[0260] The base station transmits a second DCI to which the changed  $K_{\text{2 min}}$  is applied (a second

DCI based on the changed  $K - 2$  min) in slot  $n + X$  of the scheduling cell (S194). Thereafter, the PUSCH scheduled by the second DCI is received in the scheduled cell (S196). The time interval between the second DCI and the PUSCH should be equal to or greater than the changed  $K - 2$  min. In the time interval, the UE may perform a sleep operation or a PDCCH decoding relaxation operation to save power (S195).

[0261] FIG. 20 illustrates a signaling method between a network (base station) and a UE.

[0262] Referring to FIG. 20, the base station provides a higher layer signal for setting  $K - 0$  min values to the UE (S201). For example, 'minimumSchedulingOffsetK 0' may be provided through 'PDSCH-Config' used to set UE-specific PDSCH parameters, and 'minimumSchedulingOffsetK 0' may include a list of  $K - 0$  min values.

[0263] The base station transmits a third DCI including information indicating a change to  $K - 0$  min to the UE in slot  $m$  of the scheduling cell (S202). The third DCI may be DCI format 0\_1 or DCI format 1\_1. The third DCI may be transmitted within the first three symbols of slot  $m$  or outside the first three symbols, and the  $Z$  value used to determine the application delay  $X$  may vary depending on where it is transmitted. This has already been described above. The third DCI may inform the change of  $K - 0$  min through a 1-bit field. This has already been described above.

[0264] The UE determines the application delay value  $X$  related to the time to apply the changed  $K - 0$  min (S203). As described above, the  $X$  value may be determined based on the current  $K - 0$  min of the scheduled cell, the SCS configurations of the scheduling cell and the scheduled cell, and a predetermined value dependent on the SCS configuration of the scheduling cell. For example, Equation 1 can be used.

[0265] The base station transmits a fourth DCI to which the changed  $K - 0$  min is applied (a fourth DCI based on the changed  $K - 0$  min) from after slot  $m + X$  of the scheduling cell (S204). Thereafter, the PDSCH scheduled by the fourth DCI is transmitted in the scheduled cell (S206). The time interval between the fourth DCI and the PDSCH should be equal to or greater than the changed  $K - 0$  min. In the time interval, the UE may perform a sleep operation or a PDCCH decoding relaxation operation to save power (S205).

[0266] FIG. 21 illustrates a wireless device applicable to this specification.

[0267] Referring to FIG. 21, the first wireless device 100 and the second wireless device 200 may transmit/receive wireless signals through various wireless access technologies (e.g., LTE, NR).

[0268] The first wireless device 100 includes at least one processor 102 and at least one memory 104 and may further include at least one transceiver 106 and/or at least one antenna 108. The processor 102 may be configured to control the memory 104 and/or the transceiver 106 and to implement the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed herein. For example, the processor 102 may process information in the memory 104 to generate first information/signal and may then transmit a radio signal including the first information/signal through the transceiver 106. In addition, the processor 102 may receive a radio signal including second information/signal through the transceiver 106 and may store information obtained from signal processing of the second information/signal in the memory 104. The memory 104 may be connected to the processor 102 and may store various pieces of information related to the operation of the processor 102. For example, the memory 104 may store a software code including instructions to perform some or all of processes controlled by the processor 102 or to perform the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed herein. Here, the processor 102 and the memory 104 may be part of a communication modem/circuit/chip designed to implement a radio communication technology (e.g., LTE or NR). The transceiver 106 may be connected with the processor 102 and may transmit and/or receive a radio signal via the at least one antennas 108. The transceiver 106 may include a transmitter and/or a receiver. The transceiver 106 may be replaced with a radio frequency (RF) unit. In this specification, the wireless device may refer to a communication modem/circuit/chip.

[0269] The second wireless device 200 includes at least one processor 202 and at least one memory

**204** and may further include at least one transceiver **206** and/or at least one antenna **208**. The processor **202** may be configured to control the memory **204** and/or the transceiver **206** and to implement the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed herein. For example, the processor **202** may process information in the memory **204** to generate third information/signal and may then transmit a radio signal including the third information/signal through the transceiver **206**. In addition, the processor **202** may receive a radio signal including fourth information/signal through the transceiver **206** and may store information obtained from signal processing of the fourth information/signal in the memory **204**. The memory **204** may be connected to the processor **202** and may store various pieces of information related to the operation of the processor **202**. For example, the memory **204** may store a software code including instructions to perform some or all of processes controlled by the processor **202** or to perform the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed herein. Here, the processor **202** and the memory **204** may be part of a communication modem/circuit/chip designed to implement a radio communication technology (e.g., LTE or NR). The transceiver **206** may be connected with the processor **202** and may transmit and/or receive a radio signal via the at least one antennas **208**. The transceiver **206** may include a transmitter and/or a receiver. The transceiver **206** may be replaced with an RF unit. In this specification, the wireless device may refer to a communication modem/circuit/chip.

[0270] Hereinafter, hardware elements of the wireless devices **100** and **200** are described in detail. At least one protocol layer may be implemented, but limited to, by the at least one processor **102** and **202**. For example, the at least one processor **102** and **202** may implement at least one layer (e.g., a functional layer, such as PHY, MAC, RLC, PDCP, RRC, and SDAP layers). The at least one processor **102** and **202** may generate at least one protocol data unit (PDU) and/or at least one service data unit (SDU) according to the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed herein. The at least one processor **102** and **202** may generate a message, control information, data, or information according to the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed herein. The at least one processor **102** and **202** may generate a signal (e.g., a baseband signal) including a PDU, an SDU, a message, control information, data, or information according to the functions, procedures, proposals, and/or methods disclosed herein and may provide the signal to the at least one transceiver **106** and **206**. The at least one processor **102** and **202** may receive a signal (e.g., a baseband signal) from the at least one transceiver **106** and **206** and may obtain a PDU, an SDU, a message, control information, data, or information according to the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed herein.

[0271] The at least one processor **102** and **202** may be referred to as a controller, a microcontroller, a microprocessor, or a microcomputer. The at least one processor **102** and **202** may be implemented by hardware, firmware, software, or a combination thereof. For example, at least one application-specific integrated circuit (ASIC), at least one digital signal processor (DSP), at least one digital signal processing devices (DSPD), at least one programmable logic devices (PLD), or at least one field programmable gate array (FPGA) may be included in the at least one processor **102** and **202**. The one or more processors **102** and **202** may be implemented as at least one computer readable medium (CRM) including instructions based on being executed by the at least one processor.

[0272] For example, each method described in FIGS. **16** to **20** may be performed by at least one computer readable medium (CRM) including instructions based on being executed by at least one processor. The CRM may perform, for example, receiving downlink control information (DCI) including information for a change to a value of  $K_0$  min or  $K_2$  min in a slot  $n$  of a scheduling cell, each of the  $K_0$  min and  $K_2$  min being an applied minimum scheduling offset restriction, and applying a changed  $K_0$  min or a changed  $K_2$  min value in a slot  $n+X$  of the scheduling cell. The  $X$  value is a largest value among i) a first value obtained by multiplying currently applied  $K_0$  min ( $Y$ ) in a scheduled cell scheduled by the DCI by  $2 \cdot \mu_{\text{scheduling}} / 2 \cdot \mu_{\text{scheduled}}$  and then

performing ceiling and ii) a second value (Z) that are predetermined depending on a subcarrier spacing (SCS) of the scheduling cell. The  $\mu$ scheduling is a subcarrier spacing configuration of the scheduling cell and the  $\mu$ scheduled is a subcarrier spacing configuration of the scheduled cell. [0273] The descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed herein may be implemented using firmware or software, and the firmware or software may be configured to include modules, procedures, functions, and the like. The firmware or software configured to perform the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed herein may be included in the at least one processor **102** and **202** or may be stored in the at least one memory **104** and **204** and may be executed by the at least one processor **102** and **202**. The descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed herein may be implemented in the form of a code, an instruction, and/or a set of instructions using firmware or software.

[0274] The at least one memory **104** and **204** may be connected to the at least one processor **102** and **202** and may store various forms of data, signals, messages, information, programs, codes, indications, and/or commands. The at least one memory **104** and **204** may be configured as a ROM, a RAM, an EPROM, a flash memory, a hard drive, a register, a cache memory, a computer-readable storage medium, and/or a combinations thereof. The at least one memory **104** and **204** may be disposed inside and/or outside the at least one processor **102** and **202**. In addition, the at least one memory **104** and **204** may be connected to the at least one processor **102** and **202** through various techniques, such as a wired or wireless connection.

[0275] The at least one transceiver **106** and **206** may transmit user data, control information, a radio signal/channel, or the like mentioned in the methods and/or operational flowcharts disclosed herein to at least different device. The at least one transceiver **106** and **206** may receive user data, control information, a radio signal/channel, or the like mentioned in the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed herein from at least one different device. For example, the at least one transceiver **106** and **206** may be connected to the at least one processor **102** and **202** and may transmit and receive a radio signal. For example, the at least one processor **102** and **202** may control the at least one transceiver **106** and **206** to transmit user data, control information, or a radio signal to at least one different device. In addition, the at least one processor **102** and **202** may control the at least one transceiver **106** and **206** to receive user data, control information, or a radio signal from at least one different device. The at least one transceiver **106** and **206** may be connected to the at least one antenna **108** and **208** and may be configured to transmit or receive user data, control information, a radio signal/channel, or the like mentioned in the descriptions, functions, procedures, proposals, methods, and/or operational flowcharts disclosed herein through the at least one antenna **108** and **208**. In this document, the at least one antenna may be a plurality of physical antennas or may be a plurality of logical antennas (e.g., antenna ports). The at least one transceiver **106** and **206** may convert a received radio signal/channel from an RF band signal into a baseband signal in order to process received user data, control information, a radio signal/channel, or the like using the at least one processor **102** and **202**. The at least one transceiver **106** and **206** may convert user data, control information, a radio signal/channel, or the like, processed using the at least one processor **102** and **202**, from a baseband signal to an RF bad signal. To this end, the at least one transceiver **106** and **206** may include an (analog) oscillator and/or a filter.

[0276] FIG. 22 shows an example of a structure of a signal processing module. Herein, signal processing may be performed in the processors **102** and **202** of FIG. 21.

[0277] Referring to FIG. 22, the transmitting device (e.g., a processor, the processor and a memory, or the processor and a transceiver) in a UE or BS may include a scrambler **301**, a modulator **302**, a layer mapper **303**, an antenna port mapper **304**, a resource block mapper **305**, and a signal generator **306**.

[0278] The transmitting device can transmit one or more codewords. Coded bits in each codeword

are scrambled by the corresponding scrambler **301** and transmitted over a physical channel. A codeword may be referred to as a data string and may be equivalent to a transport block which is a data block provided by the MAC layer.

[0279] Scrambled bits are modulated into complex-valued modulation symbols by the corresponding modulator **302**. The modulator **302** can modulate the scrambled bits according to a modulation scheme to arrange complex-valued modulation symbols representing positions on a signal constellation. The modulation scheme is not limited and m-PSK (m-Phase Shift Keying) or m-QAM (m-Quadrature Amplitude Modulation) may be used to modulate the coded data. The modulator may be referred to as a modulation mapper.

[0280] The complex-valued modulation symbols can be mapped to one or more transport layers by the layer mapper **303**. Complex-valued modulation symbols on each layer can be mapped by the antenna port mapper **304** for transmission on an antenna port.

[0281] Each resource block mapper **305** can map complex-valued modulation symbols with respect to each antenna port to appropriate resource elements in a virtual resource block allocated for transmission. The resource block mapper can map the virtual resource block to a physical resource block according to an appropriate mapping scheme. The resource block mapper **305** can allocate complex-valued modulation symbols with respect to each antenna port to appropriate subcarriers and multiplex the complex-valued modulation symbols according to a user.

[0282] Signal generator **306** can modulate complex-valued modulation symbols with respect to each antenna port, that is, antenna-specific symbols, according to a specific modulation scheme, for example, OFDM (Orthogonal Frequency Division Multiplexing), to generate a complex-valued time domain OFDM symbol signal. The signal generator can perform IFFT (Inverse Fast Fourier Transform) on the antenna-specific symbols, and a CP (cyclic Prefix) can be inserted into time domain symbols on which IFFT has been performed. OFDM symbols are subjected to digital-analog conversion and frequency up-conversion and then transmitted to the receiving device through each transmission antenna. The signal generator may include an IFFT module, a CP inserting unit, a digital-to-analog converter (DAC) and a frequency upconverter.

[0283] FIG. 23 shows another example of a structure of a signal processing module in a transmitting device. Herein, signal processing may be performed in a processor of a UE/BS, such as the processors **102** and **202** of FIG. 21.

[0284] Referring to FIG. 23, the transmitting device (e.g., a processor, the processor and a memory, or the processor and a transceiver) in the UE or the BS may include a scrambler **401**, a modulator **402**, a layer mapper **403**, a precoder **404**, a resource block mapper **405**, and a signal generator **406**.

[0285] The transmitting device can scramble coded bits in a codeword by the corresponding scrambler **401** and then transmit the scrambled coded bits through a physical channel.

[0286] Scrambled bits are modulated into complex-valued modulation symbols by the corresponding modulator **402**. The modulator can modulate the scrambled bits according to a predetermined modulation scheme to arrange complex-valued modulation symbols representing positions on a signal constellation. The modulation scheme is not limited and  $\pi/2$ -BPSK ( $\pi/2$ -Binary Phase Shift Keying), m-PSK (m-Phase Shift Keying) or m-QAM (m-Quadrature Amplitude Modulation) may be used to modulate the coded data.

[0287] The complex-valued modulation symbols can be mapped to one or more transport layers by the layer mapper **403**.

[0288] Complex-valued modulation symbols on each layer can be precoded by the precoder **404** for transmission on an antenna port. Here, the precoder may perform transform precoding on the complex-valued modulation symbols and then perform precoding. Alternatively, the precoder may perform precoding without performing transform precoding. The precoder **404** can process the complex-valued modulation symbols according to MIMO using multiple transmission antennas to output antenna-specific symbols and distribute the antenna-specific symbols to the corresponding resource block mapper **405**. An output  $z$  of the precoder **404** can be obtained by multiplying an

output  $y$  of the layer mapper **403** by an  $N \times M$  precoding matrix  $W$ . Here,  $N$  is the number of antenna ports and  $M$  is the number of layers.

[0289] Each resource block mapper **405** maps complex-valued modulation symbols with respect to each antenna port to appropriate resource elements in a virtual resource block allocated for transmission.

[0290] The resource block mapper **405** can allocate complex-valued modulation symbols to appropriate subcarriers and multiplex the complex-valued modulation symbols according to a user.

[0291] Signal generator **406** can modulate complex-valued modulation symbols according to a specific modulation scheme, for example, OFDM, to generate a complex-valued time domain OFDM symbol signal. The signal generator **406** can perform IFFT (Inverse Fast Fourier Transform) on antenna-specific symbols, and a CP (cyclic Prefix) can be inserted into time domain symbols on which IFFT has been performed. OFDM symbols are subjected to digital-analog conversion and frequency up-conversion and then transmitted to the receiving device through each transmission antenna. The signal generator **406** may include an IFFT module, a CP inserting unit, a digital-to-analog converter (DAC) and a frequency upconverter.

[0292] The signal processing procedure of the receiving device may be reverse to the signal processing procedure of the transmitting device. Specifically, the processor of the transmitting device decodes and demodulates RF signals received through antenna ports of the transceiver. The receiving device may include a plurality of reception antennas, and signals received through the reception antennas are restored to baseband signals, and then multiplexed and demodulated according to MIMO to be restored to a data string intended to be transmitted by the transmitting device. The receiving device may include a signal restoration unit that restores received signals to baseband signals, a multiplexer for combining and multiplexing received signals, and a channel demodulator for demodulating multiplexed signal strings into corresponding codewords. The signal restoration unit, the multiplexer and the channel demodulator may be configured as an integrated module or independent modules for executing functions thereof. More specifically, the signal restoration unit may include an analog-to-digital converter (ADC) for converting an analog signal into a digital signal, a CP removal unit that removes a CP from the digital signal, an FFT module for applying FFT (fast Fourier transform) to the signal from which the CP has been removed to output frequency domain symbols, and a resource element demapper/equalizer for restoring the frequency domain symbols to antenna-specific symbols. The antenna-specific symbols are restored to transport layers by the multiplexer and the transport layers are restored by the channel demodulator to codewords intended to be transmitted by the transmitting device.

[0293] FIG. **24** illustrates an example of a wireless communication device according to an implementation example of the present disclosure.

[0294] Referring to FIG. **24**, the wireless communication device, for example, a UE may include at least one of a processor **2310** such as a digital signal processor (DSP) or a microprocessor, a transceiver **2335**, a power management module **2305**, an antenna **2340**, a battery **2355**, a display **2315**, a keypad **2320**, a global positioning system (GPS) chip **2360**, a sensor **2365**, a memory **2330**, a subscriber identification module (SIM) card **2325**, a speaker **2345** and a microphone **2350**. A plurality of antennas and a plurality of processors may be provided.

[0295] The processor **2310** can implement functions, procedures and methods described in the present description. The processor **2310** in FIG. **24** may be the processors **102** and **202** in FIG. **21**.

[0296] The memory **2330** is connected to the processor **2310** and stores information related to operations of the processor. The memory may be located inside or outside the processor and connected to the processor through various techniques such as wired connection and wireless connection. The memory **2330** in FIG. **24** may be the memories **104** and **204** in FIG. **21**.

[0297] A user can input various types of information such as telephone numbers using various techniques such as pressing buttons of the keypad **2320** or activating sound using the microphone **2350**. The processor **2310** can receive and process user information and execute an appropriate

function such as calling using an input telephone number. In some scenarios, data can be retrieved from the SIM card **2325** or the memory **2330** to execute appropriate functions. In some scenarios, the processor **2310** can display various types of information and data on the display **2315** for user convenience.

[0298] The transceiver **2335** is connected to the processor **2310** and transmit and/or receive RF signals. The processor can control the transceiver in order to start communication or to transmit RF signals including various types of information or data such as voice communication data. The transceiver includes a transmitter and a receiver for transmitting and receiving RF signals. The antenna **2340** can facilitate transmission and reception of RF signals. In some implementation examples, when the transceiver receives an RF signal, the transceiver can forward and convert the signal into a baseband frequency for processing performed by the processor. The signal can be processed through various techniques such as converting into audible or readable information to be output through the speaker **2345**. The transceiver in FIG. **24** may be the transceivers **106** and **206** in FIG. **21**.

[0299] Although not shown in FIG. **24**, various components such as a camera and a universal serial bus (USB) port may be additionally included in the UE. For example, the camera may be connected to the processor **2310**.

[0300] FIG. **24** is an example of implementation with respect to the UE and implementation examples of the present disclosure are not limited thereto. The UE need not essentially include all the components shown in FIG. **24**. That is, some of the components, for example, the keypad **2320**, the GPS chip **2360**, the sensor **2365** and the SIM card **2325** may not be essential components. In this case, they may not be included in the UE.

[0301] FIG. **25** shows an example of a processor **2000**.

[0302] Referring to FIG. **25**, the processor **2000** may include a control channel monitoring unit **2010** and a data channel receiving unit **2020**. The processor **2000** may execute the methods (the position of the receiver, for example, the position of the UE) described with reference to FIGS. **16** to **20**. For example, the processor **2000** receives downlink control information (DCI) including information notifying a change in the value of  $K_0$  min or  $K_{sub.2min}$  in slot  $n$  of a scheduling cell, each of  $K_0$  min and  $K_2$  min being an applied minimum scheduling offset restriction. Also, the processor **2000** applies the changed  $K_0$  min or the changed  $K_2$  min value in the slot  $n+X$  of the scheduling cell. The  $X$  value is a largest value among i) a first value obtained by multiplying currently applied  $K_0$  min ( $Y$ ) in a scheduled cell scheduled by the DCI by  $2 \cdot \mu_{scheduling} / 2 \cdot \mu_{scheduled}$  and then performing ceiling and ii) a second value ( $Z$ ) that are predetermined depending on a subcarrier spacing (SCS) of the scheduling cell. The  $\mu_{scheduling}$  is a subcarrier spacing configuration of the scheduling cell and the  $\mu_{scheduled}$  is a subcarrier spacing configuration of the scheduled cell. The processor **2000** may be an example of the processors **102** and **202** of FIG. **21**.

[0303] FIG. **26** shows an example of a processor **3000**.

[0304] Referring to FIG. **26**, the processor **3000** may include a control information/data generation module **3010** and a transmission module **3020**. The processor **3000** may execute the methods described from the perspective of the transmitter in FIGS. **16** to **20**. For example, the processor **3000** transmits, to a user equipment, downlink control information (DCI) including information for a change to a value of  $K_0$  min or  $K_2$  min in a slot  $n$  of a scheduling cell, each of the  $K_0$  min and  $K_2$  min being an applied minimum scheduling offset restriction. The processor **3000** may assume that the changed  $K_0$  min or the changed  $K_2$  min value is applied in the slot  $n+X$  of the scheduling cell. The  $X$  value is a largest value among i) a first value obtained by multiplying currently applied  $K_0$  min ( $Y$ ) in a scheduled cell scheduled by the DCI by  $2 \cdot \mu_{scheduling} / 2 \cdot \mu_{scheduled}$  and then performing ceiling and ii) a second value ( $Z$ ) that are predetermined depending on a subcarrier spacing (SCS) of the scheduling cell. The  $\mu_{scheduling}$  is a subcarrier spacing configuration of the scheduling cell and the  $\mu_{scheduled}$  is a subcarrier spacing configuration of the scheduled cell. The

processor **3000** may be an example of the processors **102** and **202** of FIG. **21**.

[0305] FIG. **27** shows another example of a wireless device.

[0306] According to FIG. **27**, a wireless device may include at least one processor **102**, **202**, at least one memory **104**, **204**, at least one transceiver **106**, **206**, and one or more antennas **108**, **208**.

[0307] The example of the wireless device described in FIG. **27** is different from the example of the wireless described in FIG. **21** in that the processors **102** and **202** and the memories **104** and **204** are separated in FIG. **21** whereas the memories **104** and **204** are included in the processors **102** and **202** in the example of FIG. **27**. That is, the processor and the memory may constitute one chipset.

[0308] FIG. **28** shows another example of a wireless device applied to the present specification.

The wireless device may be implemented in various forms according to a use-case/service.

[0309] Referring to FIG. **28**, wireless devices **100** and **200** may correspond to the wireless devices of FIG. **21** and may be configured by various elements, components, units/portions, and/or modules. For example, each of the wireless devices **100** and **200** may include a communication unit **110**, a control unit **120**, a memory unit **130**, and additional components **140**. The communication unit may include a communication circuit **112** and transceiver(s) **114**. For example, the communication circuit **112** may include the one or more processors **102** and **202** and/or the one or more memories **104** and **204**. For example, the transceiver(s) **114** may include the one or more transceivers **106** and **206** and/or the one or more antennas **108** and **208** of FIG. **21**. The control unit **120** is electrically connected to the communication unit **110**, the memory **130**, and the additional components **140** and controls overall operation of the wireless devices. For example, the control unit **120** may control an electric/mechanical operation of the wireless device based on programs/code/commands/information stored in the memory unit **130**. In addition, the control unit **120** may transmit the information stored in the memory unit **130** to the exterior (e.g., other communication devices) via the communication unit **110** through a wireless/wired interface or store, in the memory unit **130**, information received through the wireless/wired interface from the exterior (e.g., other communication devices) via the communication unit **110**.

[0310] The additional components **140** may be variously configured according to types of wireless devices. For example, the additional components **140** may include at least one of a power unit/battery, input/output (I/O) unit, a driving unit, and a computing unit. The wireless device may be implemented in the form of, without being limited to, the robot (**100a** of FIG. **30**), the vehicles (**100b-1** and **100b-2** of FIG. **30**), the XR device (**100c** of FIG. **30**), the hand-held device (**100d** of FIG. **30**), the home appliance (**100e** of FIG. **30**), the IoT device (**100f** of FIG. **30**), a digital broadcast UE, a hologram device, a public safety device, an MTC device, a medicine device, a fintech device (or a finance device), a security device, a climate/environment device, the AI server/device (**400** of FIG. **30**), the BSs (**200** of FIG. **30**), a network node, etc. The wireless device may be used in a mobile or fixed place according to a use-example/service.

[0311] In FIG. **28**, the entirety of the various elements, components, units/portions, and/or modules in the wireless devices **100** and **200** may be connected to each other through a wired interface or at least a part thereof may be wirelessly connected through the communication unit **110**. For example, in each of the wireless devices **100** and **200**, the control unit **120** and the communication unit **110** may be connected by wire and the control unit **120** and first units (e.g., **130** and **140**) may be wirelessly connected through the communication unit **110**. In addition, each element, component, unit/portion, and/or module within the wireless devices **100** and **200** may further include one or more elements. For example, the control unit **120** may be configured by a set of one or more processors. For example, the control unit **120** may be configured by a set of a communication control processor, an application processor, an Electronic Control Unit (ECU), a graphical processing unit, and a memory control processor. For another example, the memory **130** may be configured by a Random Access Memory (RAM), a Dynamic RAM (DRAM), a Read Only Memory (ROM), a flash memory, a volatile memory, a non-volatile memory, and/or a combination thereof.

[0312] FIG. 29 illustrates a hand-held device applied to the present specification. The hand-held device may include a smartphone, a smartpad, a wearable device (e.g., a smartwatch or a smartglasses), or a portable computer (e.g., a notebook). The hand-held device may be referred to as a mobile station (MS), a user terminal (UT), a Mobile Subscriber Station (MSS), a Subscriber Station (SS), an Advanced Mobile Station (AMS), or a Wireless Terminal (WT).

[0313] Referring to FIG. 29, a hand-held device **100** may include an antenna unit **108**, a communication unit **110**, a control unit **120**, a memory unit **130**, a power supply unit **140a**, an interface unit **140b**, and an I/O unit **140c**. The antenna unit **108** may be configured as a part of the communication unit **110**. Blocks **110** to **130/140a** to **140c** respective correspond to the blocks **110** to **130/140** of FIG. 28.

[0314] The communication unit **110** may transmit and receive signals (e.g., data and control signals) to and from other wireless devices or BSs. The control unit **120** may perform various operations by controlling constituent elements of the hand-held device **100**. The control unit **120** may include an Application Processor (AP). The memory unit **130** may store data/parameters/programs/code/commands needed to drive the hand-held device **100**. In addition, the memory unit **130** may store input/output data/information. The power supply unit **140a** may supply power to the hand-held device **100** and include a wired/wireless charging circuit, a battery, etc. The interface unit **140b** may support connection of the hand-held device **100** to other external devices. The interface unit **140b** may include various ports (e.g., an audio I/O port and a video I/O port) for connection with external devices. The I/O unit **140c** may input or output video information/signals, audio information/signals, data, and/or information input by a user. The I/O unit **140c** may include a camera, a microphone, a user input unit, a display unit **140d**, a speaker, and/or a haptic module.

[0315] For example, in the case of data communication, the I/O unit **140c** may acquire information/signals (e.g., touch, text, voice, images, or video) input by a user and the acquired information/signals may be stored in the memory unit **130**. The communication unit **110** may convert the information/signals stored in the memory into radio signals and transmit the converted radio signals to other wireless devices directly or to a BS. In addition, the communication unit **110** may receive radio signals from other wireless devices or the BS and then restore the received radio signals into original information/signals. The restored information/signals may be stored in the memory unit **130** and may be output as various types (e.g., text, voice, images, video, or haptic) through the I/O unit **140c**.

[0316] FIG. 30 illustrates a communication system **1** applied to the present specification.

[0317] Referring to FIG. 30, a communication system **1** applied to the present specification includes wireless devices, Base Stations (BSs), and a network. Herein, the wireless devices represent devices performing communication using Radio Access Technology (RAT) (e.g., 5G New RAT (NR)) or Long-Term Evolution (LTE)) and may be referred to as communication/radio/5G devices. The wireless devices may include, without being limited to, a robot **100a**, vehicles **100b-1** and **100b-2**, an extended Reality (X R) device **100c**, a hand-held device **100d**, a home appliance **100e**, an Internet of Things (IoT) device **100f**, and an Artificial Intelligence (AI) device/server **400**. For example, the vehicles may include a vehicle having a wireless communication function, an autonomous vehicle, and a vehicle capable of performing communication between vehicles. Herein, the vehicles may include an Unmanned Aerial Vehicle (UAV) (e.g., a drone). The X R device may include an Augmented Reality (AR)/Virtual Reality (VR)/Mixed Reality (MR) device and may be implemented in the form of a Head-Mounted Device (HMD), a Head-Up Display (HUD) mounted in a vehicle, a television, a smartphone, a computer, a wearable device, a home appliance device, a digital signage, a vehicle, a robot, etc. The hand-held device may include a smartphone, a smartpad, a wearable device (e.g., a smartwatch or a smartglasses), and a computer (e.g., a notebook). The home appliance may include a TV, a refrigerator, and a washing machine. The IoT device may include a sensor and a smartmeter. For example, the BSs and the network may

be implemented as wireless devices and a specific wireless device **200a** may operate as a BS/network node with respect to other wireless devices.

[0318] The wireless devices **100a** to **100f** may be connected to the network **300** via the B Ss **200**. An AI technology may be applied to the wireless devices **100a** to **100f** and the wireless devices **100a** to **100f** may be connected to the AI server **400** via the network **300**. The network **300** may be configured using a 3G network, a 4G (e.g., LTE) network, or a 5G (e.g., NR) network. Although the wireless devices **100a** to **100f** may communicate with each other through the B Ss **200**/network **300**, the wireless devices **100a** to **100f** may perform direct communication (e.g., sidelink communication) with each other without passing through the BSs/network. For example, the vehicles **100b-1** and **100b-2** may perform direct communication (e.g. Vehicle-to-Vehicle (V2V)/Vehicle-to-everything (V2X) communication). In addition, the IoT device (e.g., a sensor) may perform direct communication with other IoT devices (e.g., sensors) or other wireless devices **100a** to **100f**.

[0319] Wireless communication/connections **150a**, **150b**, or **150c** may be established between the wireless devices **100a** to **100f**/BS **200**, or BS **200**/BS **200**. Herein, the wireless communication/connections may be established through various RATs (e.g., 5G NR) such as uplink/downlink communication **150a**, sidelink communication **150b** (or, D2D communication), or inter BS communication (e.g. relay, Integrated Access Backhaul (IAB)). The wireless devices and the B Ss/the wireless devices may transmit/receive radio signals to/from each other through the wireless communication/connections **150a** and **150b**. For example, the wireless communication/connections **150a** and **150b** may transmit/receive signals through various physical channels. To this end, at least a part of various configuration information configuring processes, various signal processing processes (e.g., channel encoding/decoding, modulation/demodulation, and resource mapping/demapping), and resource allocating processes, for transmitting/receiving radio signals, may be performed based on the various proposals of the present disclosure.

[0320] Here, the wireless communication technology implemented in the wireless devices **100** and **200** of the present specification may include a narrowband Internet of Things for low-power communication as well as LTE, NR, and 6G. At this time, for example, NB-IoT technology may be an example of LPWAN (Low Power Wide Area Network) technology, and it may be implemented in standards such as LTE Cat NB 1 and/or LTE Cat NB2, and is not limited to the above-described name. Additionally or alternatively, the wireless communication technology implemented in the wireless devices **100** and **200** of the present specification may perform communication based on the LTE-M technology. In this case, as an example, the LTE-M technology may be an example of an LPWAN technology, and may be called by various names such as enhanced machine type communication (eMTC). For example, LTE-M technology may be implemented in at least one of various standards such as 1) LTE CAT 0, 2) LTE Cat M1, 3) LTE Cat M2, 4) LTE non-BL (non-Bandwidth Limited), 5) LTE-MTC, 6) LTE Machine Type Communication, and/or 7) LTE M, and is not limited to the above-described name. Additionally or alternatively, the wireless communication technology implemented in the wireless devices **100** and **200** of the present specification may include at least one of ZigBee, Bluetooth, and Low Power Wide Area Network (LPWAN) in consideration of low power communication, and is not limited to the above-mentioned names. For example, the ZigBee technology can create PAN (personal area networks) related to small/low-power digital communication based on various standards such as IEEE 802.15.4, and can be called by various names.

[0321] NR supports a plurality of numerologies (or a plurality of ranges of subcarrier spacing (SCS)) in order to support a variety of 5G services. For example, when SCS is 15 kHz, a wide area in traditional cellular bands is supported; when SCS is 30 kHz/60 kHz, a dense-urban, lower-latency, and wider-carrier bandwidth is supported; when SCS is 60 KHz or higher, a bandwidth greater than 24.25 GHz is supported to overcome phase noise.

[0322] NR frequency bands may be defined as frequency ranges of two types (FR 1 and FR2). The

values of the frequency ranges may be changed. For example, the frequency ranges of the two types (FR 1 and FR 2) may be as shown in Table 7. For convenience of description, FR1 of the frequency ranges used for an NR system may refer to a “sub 6 GHz range”, and FR2 may refer to an “above 6 GHz range” and may be referred to as a millimeter wave (mmW).

TABLE-US-00008 TABLE 7 Frequency Corresponding range frequency Subcarrier designation range spacing FR1 450 MHz-6000 MHz 15, 30, 60 kHz FR2 24250 MHz-52600 MHz 60, 120, 240 kHz

[0323] As illustrated above, the values of the frequency ranges for the NR system may be changed. For example, FR 1 may include a band from 410 MHz to 7125 MHz as shown in Table 8. That is, FR 1 may include a frequency band of 6 GHz (or 5850, 5900, 5925 MHz, or the like) or greater. For example, the frequency band of 6 GHz (or 5850, 5900, 5925 MHz, or the like) or greater included in FR 1 may include an unlicensed band. The unlicensed bands may be used for a variety of purposes, for example, for vehicular communication (e.g., autonomous driving).

TABLE-US-00009 TABLE 8 Frequency Corresponding range frequency Subcarrier designation range spacing FR1 410 MHz-7125 MHz 15, 30, 60 kHz FR2 24250 MHz-52600 MHz 60, 120, 240 kHz

[0324] FIG. 31 illustrates a vehicle or an autonomous driving vehicle applied to this specification. The vehicle or the autonomous driving vehicle may be configured as a mobile robot, a car, a train, a manned/unmanned aerial vehicle (AV), a ship, or the like.

[0325] Referring to FIG. 31, the vehicle or the autonomous driving vehicle 100 may include an antenna unit 108, a communication unit 110, a control unit 120, a driving unit 140a, a power supply unit 140b, a sensor unit 140c, and an autonomous driving unit 140d. The antenna unit 108 may be configured as a part of the communication unit 110. Blocks 110/130/140a to 140d correspond to the blocks 110/130/140 in FIG. 28, respectively.

[0326] The communication unit 110 may transmit and receive a signal (e.g., data, a control signal, or the like) to and from external devices, such as a different vehicle, a base station (e.g. a base station, a road-side unit, or the like), and a server. The control unit 120 may control elements of the vehicle or the autonomous driving vehicle 100 to perform various operations. The control unit 120 may include an electronic control unit (ECU). The driving unit 140a may enable the vehicle or the autonomous driving vehicle 100 to run on the ground. The driving unit 140a may include an engine, a motor, a power train, wheels, a brake, a steering device, and the like. The power supply unit 140b supplies power to the vehicle or the autonomous driving vehicle 100 and may include a wired/wireless charging circuit, a battery, and the like. The sensor unit 140c may obtain a vehicle condition, environmental information, user information, and the like. The sensor unit 140c may include an inertial measurement unit (IM U) sensor, a collision sensor, a wheel sensor, a speed sensor, an inclination sensor, a weight sensor, a heading sensor, a position module, vehicular forward/backward vision sensors, a battery sensor, a fuel sensor, a tire sensor, a steering sensor, a temperature sensor, a humidity sensor, an ultrasonic sensor, an illuminance sensor, a pedal position sensor, and the like. The autonomous driving unit 140d may implement a technology for maintaining a driving lane, a technology for automatically adjusting speed, such as adaptive cruise control, a technology for automatic driving along a set route, a technology for automatically setting a route and driving when a destination is set, and the like.

[0327] For example, the communication unit 110 may receive map data, traffic condition data, and the like from an external server. The autonomous driving unit 140d may generate an autonomous driving route and a driving plan on the basis of obtained data. The control unit 120 may control the driving unit 140a to move the vehicle or the autonomous driving vehicle 100 along the autonomous driving route according to the driving plan (e.g., speed/direction control). During autonomous driving, the communication unit 110 may aperiodically/periodically obtain updated traffic condition data from the external server and may obtain surrounding traffic condition data from a neighboring vehicle. Further, during autonomous driving, the sensor unit 140c may obtain a vehicle

condition and environmental information. The autonomous driving unit **140d** may update the autonomous driving route and the driving plan on the basis of newly obtained data/information. The communication unit **110** may transmit information about a vehicle location, an autonomous driving route, a driving plan, and the like to the external server. The external server may predict traffic condition data in advance using AI technology or the like on the basis of information collected from vehicles or autonomous driving vehicles and may provide the predicted traffic condition data to the vehicles or the autonomous driving vehicles.

[0328] Claims disclosed in the present specification can be combined in various ways. For example, technical features in method claims of the present specification can be combined to be implemented or performed in an apparatus, and technical features in apparatus claims of the present specification can be combined to be implemented or performed in a method. Further, technical features in method claims and apparatus claims of the present specification can be combined to be implemented or performed in an apparatus. Further, technical features in method claims and apparatus claims of the present specification can be combined to be implemented or performed in a method.

## Claims

1. A method comprising: receiving, by a user equipment (UE), first downlink control information (DCI) with a field for a change of  $K_{\text{sub.2min}}$  within first three symbols of a slot  $n$  of a scheduling cell, wherein the  $K_{\text{sub.2min}}$  is an applied minimum scheduling offset restriction related to a minimum slot offset between a slot for receiving DCI and a slot for transmitting a physical uplink shared channel (PUSCH) scheduled by the received DCI, and applying, by the UE, a changed  $K_{\text{sub.2min}}$  in slot  $n+X$  of the scheduling cell, wherein the  $X$  is determined based on a following equation,  $X = \max(\lceil \frac{2 \cdot \mu_{\text{sub.PDCCH}}}{\mu_{\text{sub.PDSCH}}} \rceil, Z)$  wherein the  $Y$  is a currently applied  $K_{\text{sub.0min}}$  value in a scheduled cell, the  $K_{\text{sub.0min}}$  is an applied minimum scheduling offset restriction related to a minimum slot offset between a slot for receiving specific DCI and a slot for receiving a physical downlink shared channel (PDSCH) scheduled by the specific DCI, the  $\mu_{\text{sub.PDCCH}}$  is a subcarrier spacing configuration for a physical downlink control channel (PDCCH) in the scheduling cell, the  $\mu_{\text{sub.PDSCH}}$  is a subcarrier spacing configuration for a PDSCH in the scheduled cell, and the  $Z$  is determined based on a subcarrier spacing in the scheduling cell, wherein based on the subcarrier spacing in the scheduling cell being 15 kHz, 30 KHz, 60 kHz, 120 kHz, the  $Z$  is 1, 1, 2, 2, respectively, and wherein, based on that the first DCI is received outside the first three symbols of the slot  $n$ , the  $Z$  is incremented by one before determining the  $X$ .
2. The method of claim 1, further comprising: receiving, by the UE, second DCI based on the changed  $K_{\text{sub.2min}}$ ; and performing, by the UE, a PUSCH transmission scheduled by the second DCI.
3. The method of claim 1, further comprising: receiving, by the UE, third DCI with a field for a change of  $K_{\text{sub.0min}}$  within first three symbols of slot  $m$  of the scheduling cell, applying, by the UE, a changed  $K_{\text{sub.0min}}$  in slot  $m+X$  of the scheduling cell.
4. The method of claim 3, wherein based on that the third DCI is received outside the first three symbols of the slot  $m$ , the  $Z$  is incremented by one before determining the  $X$ .
5. The method of claim 3, further comprising: receiving, by the UE, fourth DCI based on the changed  $K_{\text{sub.0min}}$ ; and performing, by the UE, a PDSCH reception scheduled by the fourth DCI.
6. The method of claim 1, wherein the slot  $n$  includes 14 symbols in a time domain.
7. A user equipment (UE) comprising: at least one memory; at least one transceiver for transmitting and receiving a radio signal; and at least one processor operably connectable to the at least one transceiver and the at least one memory, wherein the at least one memory stores instructions that,

based on being executed by the at least one processor, cause the at least one processor to perform operations comprising: receiving first downlink control information (DCI) with a field for a change of  $K_{sub.2min}$  within first three symbols of a slot  $n$  of a scheduling cell, wherein the  $K_{sub.2min}$  is an applied minimum scheduling offset restriction related to a minimum slot offset between a slot for receiving DCI and a slot for transmitting a physical uplink shared channel (PUSCH) scheduled by the received DCI, and applying a changed  $K_{sub.2min}$  in slot  $n+X$  of the scheduling cell, wherein the  $X$  is determined based on a following equation,  $X = \max(\text{Math. } Y \cdot \text{Math. } \frac{2}{2} \frac{PDCCH}{PDSCH} \cdot \text{Math. } , Z)$  wherein the  $Y$  is a currently applied  $K_{sub.0min}$  value in a scheduled cell, the  $K_{sub.0min}$  is an applied minimum scheduling offset restriction related to a minimum slot offset between a slot for receiving specific DCI and a slot for receiving a physical downlink shared channel (PDSCH) scheduled by the specific DCI, the  $\mu_{sub.PDCCH}$  is a subcarrier spacing configuration for a physical downlink control channel (PDCCH) in the scheduling cell, the  $\mu_{sub.PDSCH}$  is a subcarrier spacing configuration for a PDSCH in the scheduled cell, and the  $Z$  is determined based on a subcarrier spacing in the scheduling cell, wherein based on the subcarrier spacing in the scheduling cell being 15 kHz, 30 kHz, 60 kHz, 120 kHz, the  $Z$  is 1, 1, 2, 2, respectively, and wherein, based on that the first DCI is received outside the first three symbols of the slot  $n$ , the  $Z$  is incremented by one before determining the  $X$ .

**8.** The UE of claim 7, the operations further comprising: receiving second DCI based on the changed  $K_{sub.2min}$ ; and performing a PUSCH transmission scheduled by the second DCI.

**9.** The UE of claim 7, the operations further comprising: receiving third DCI with a field for a change of  $K_{sub.0min}$  within first three symbols of slot  $m$  of the scheduling cell, applying a changed  $K_{sub.0min}$  in slot  $m+X$  of the scheduling cell.

**10.** The UE of claim 9, wherein based on that the third DCI is received outside the first three symbols of the slot  $m$ , the  $Z$  is incremented by one before determining the  $X$ .

**11.** The UE of claim 9, the operations further comprising: receiving fourth DCI based on the changed  $K_{sub.0min}$ ; and performing a PDSCH reception scheduled by the fourth DCI.

**12.** The UE of claim 7, wherein the slot  $n$  includes 14 symbols in a time domain.

**13.** A base station (BS) comprising: at least one memory; at least one transceiver for transmitting and receiving a radio signal; and at least one processor operably connectable to the at least one transceiver and the at least one memory, wherein the at least one memory stores instructions that, based on being executed by the at least one processor, cause the at least one processor to perform operations comprising: transmitting, to a user equipment (UE), first downlink control information (DCI) with a field for a change of  $K_{sub.2min}$  within first three symbols of a slot  $n$  of a scheduling cell, wherein the  $K_{sub.2min}$  is an applied minimum scheduling offset restriction related to a minimum slot offset between a slot for transmitting DCI and a slot for receiving a physical uplink shared channel (PUSCH) scheduled by the received DCI, and applying a changed  $K_{sub.2min}$  in slot  $n+X$  of the scheduling cell, wherein the  $X$  is determined based on a following equation,  $X = \max(\text{Math. } Y \cdot \text{Math. } \frac{2}{2} \frac{PDCCH}{PDSCH} \cdot \text{Math. } , Z)$  wherein the  $Y$  is a currently applied  $K_{sub.0min}$  value in a scheduled cell, the  $K_{sub.0min}$  is an applied minimum scheduling offset restriction related to a minimum slot offset between a slot for transmitting specific DCI and a slot for transmitting a physical downlink shared channel (PDSCH) scheduled by the specific DCI, the  $\mu_{sub.PDCCH}$  is a subcarrier spacing configuration for a physical downlink control channel (PDCCH) in the scheduling cell, the  $\mu_{sub.PDSCH}$  is a subcarrier spacing configuration for a PDSCH in the scheduled cell, and the  $Z$  is determined based on a subcarrier spacing in the scheduling cell, wherein based on the subcarrier spacing in the scheduling cell being 15 kHz, 30 kHz, 60 kHz, 120 kHz, the  $Z$  is 1, 1, 2, 2, respectively, and wherein, based on that the first DCI is transmitted outside the first three symbols of the slot  $n$ , the  $Z$  is incremented by one before determining the  $X$ .